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 SURVEY

Machine Learning and Deep Learning Approaches for Fake News Detection: A Systematic Review of Techniques, Challenges, and Advancements

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ABSTRACT In response to the escalating threat of fake news on social media, this systematic literature review analyzes the recent advancements in machine learning and deep learning approaches for automated detection. Following the PRISMA guidelines, we examined 90 peer-reviewed studies published between 2020 and 2024 to evaluate the model effectiveness, identify limitations, and highlight emerging trends. Our analysis shows that deep learning models, particularly transformer-based architectures such as BERT, consistently outperform traditional machine learning methods, often achieving a high accuracy (Acc), precision (P), recall (R), and F1-score (F1). For instance, a BERT-based model reported up to 99.9% accuracy on the Kaggle fake news dataset and above 98% accuracy on other public datasets, including ISOT, Fake-or-Real, and D3. Similarly, the GANM model demonstrated robust performance on the FakeNewsNet dataset by integrating text and social features. Transfer learning and multimodal models that incorporate user behaviour and network information significantly improve detection in diverse, low-resource environments. However, challenges persist in terms of the dataset quality, model interpretability, domain generalisability, and real-time deployment. This review also underscores the limited adoption of few-shot and zero-shot learning techniques, highlighting a promising direction for future research on handling emerging misinformation using minimal training data. To support practical deployment, we advocate the development of explainable, multilingual, and lightweight models with greater emphasis on human-centred evaluation and ethical considerations. Our findings provide a foundation for researchers and practitioners to build scalable, trustworthy, and context-aware fake news detection systems for global use.

INDEX TERMS Fake news detection, machine learning, deep learning, natural language processing, text analysis, text classification, sentiment analysis, early detection.

I. INTRODUCTION

The proliferation of fake news in the digital age has emerged as a critical issue, with significant implications for society, politics, and public discourse. In this review, we explicitly define “fake news” as intentionally fabricated, false, or misleading information created and disseminated with

the deliberate aim of deceiving readers or influencing public opinion. Notably, this definition excludes satire, parody, and other humorous content, which, despite their potential for misinterpretation, lacks the intent to deceive. It also excludes unintentionally incorrect information, such as genuine journalistic errors or honest misunderstandings. The rapid dissemination of fake news is particularly pronounced on social media platforms because of their expansive user base and inadequate content verification mechanisms. [1],

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[2], [3], [4]. The impact of fake news is profound, ranging from eroding trust in media and public institutions to swaying elections, exacerbating social divisions, and even inciting violence [5], [6], [7]. In the healthcare domain, fake news has led to increased vaccine hesitancy and the spread of dangerous misinformation, particularly during the COVID-19 pandemic [8], [9], [10], [11]. Given these multifaceted consequences, there is an urgent need for effective methods for detecting and mitigating the spread of fake news.

In response to this challenge, researchers have turned to machine learning and deep learning approaches as promising tools for automated fake news detection. These techniques leverage various news articles and their dissemination, including linguistic patterns, visual cues, user interactions, and propagation characteristics [12], [13], [14]. Traditional machine learning models such as support vector machines, random forests, and gradient boosting have been widely applied [15], [16], [17]. However, deep learning architectures have gained significant traction because of their superior performance in capturing complex patterns. Recent literature shows a growing shift towards advanced deep learning models, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and transformer architectures such as BERT, which have demonstrated high effectiveness across fake news detection tasks [18], [19], [20], [21], [22], [23], [24], [25]. These models capture contextual information and subtle linguistic nuances that may elude traditional approaches to language processing. Another emerging direction is the application of graph neural networks to model the propagation patterns of news articles through social networks, enabling the incorporation of user interaction data into detection systems [26], [27], [28]. Researchers have also explored multimodal approaches that combine textual, visual, and social context information to improve detection accuracy [29], [30], [31]. Additionally, there is growing interest in developing explainable AI (XAI) systems that not only classify news as fake or real, but also provide interpretable rationales for their decisions [32], [33], [34].

Despite the promising results achieved by machine learning and deep learning models, several challenges remain in the field of automated fake news detection. These include the need for large, high-quality labelled datasets, difficulty in generalising models across different domains and languages, and adversarial nature of fake news creation [35], [36], [37], [38]. The dynamic and evolving nature of fake news poses a particular challenge, as detection models must adapt to new types of misinformation and disinformation tactics [39], [40], [41]. Furthermore, there are ongoing discussions about the ethical implications of automated content moderation and the potential for false positives to infringe on free speech [42], [43], [44]. As research in this area continues to evolve, future work will likely focus on developing more robust, interpretable, and context-aware models that can adapt to the

dynamic landscape of online misinformation while addressing these critical challenges [45], [46], [47].

This systematic literature review (SLR) aims to critically analyse the current landscape of text-based fake news detection methods on social media platforms with a specific focus on machine learning and deep learning approaches. The primary contributions of this study are multi-faceted.

- This study investigated effective algorithms and feature extraction techniques for fake news classification by comparing traditional and deep learning models in terms of accuracy, efficiency, and generalisability.
- The integration of contextual information and user meta-data was examined to improve the detection accuracy and enhance models beyond content-only analysis.
- The scalability and real-time performance of current detection systems were assessed to handle the growing volume and velocity of social media data.
- Transfer learning and multimodal approaches have been explored, adapting pre-trained models for fake news detection and combining text with images and behavioural data.
- The efficacy of few-shot and zero-shot learning techniques in detecting emerging misinformation was examined, offering the potential for enhanced detection with minimal training data.

Through these contributions, this review aims to provide a systematic understanding of the current state of fake news detection research, identify key challenges, and highlight promising directions for future research. The remainder of this paper is organised as follows. Section II presents related studies. Section III introduces the methodology adopted for the systematic literature review. Section IV presents the analysis based on research questions, explicates the challenges and gaps, and demonstrates opportunities in the area. Section V discusses the limitations of this study. Finally, Section VI concludes the study.

II. RELATED WORK

The field of fake news detection has gained significant attention in recent years, driven by the increasing prevalence of misinformation on social media platforms [48], [49], [50]. This review examines the current state of research on automated fake news detection, focusing on machine learning (ML) and natural language processing (NLP) approaches [51], [52], [53]. The evolution of fake news detection techniques has reflected the complexity of this problem. Early approaches relied on traditional ML classifiers and basic NLP techniques [3], [54]. However, the field has rapidly progressed towards more sophisticated methods, particularly deep learning models [5], [49], [52]. Transformer-based architectures, such as BERT and RoBERTa, have become central in this field due to their superior contextual representation capabilities [49], [50], [53], [55], driving a shift away from traditional ML methods. These models capture contextual and semantic relationships within text, making

them well suited for distinguishing between genuine and fabricated news articles.

Researchers have also recognised the multifaceted nature of fake news, leading to the development of multimodal approaches [2], [51]. These methods combine textual analysis with visual and structural data to provide a more comprehensive understanding of the misinformation. In addition, graph-based techniques and social network analysis have been employed to model the propagation patterns of fake news, acknowledging the crucial role of social networks in the spread of misinformation [2], [5]. Despite these advancements, several challenges remain to be overcome. The need for large high-quality datasets for training remains a major hurdle [1], [48], [53], [56]. The dynamic nature of misinformation tactics further complicates detection, as models must quickly adapt to new formats and strategies. Generalising detection systems across different domains, languages, and cultural contexts has proven difficult, spurring interest in transfer learning and multilingual techniques [50], [54], [57]. Explainability and interpretability have become essential aspects of modern ML systems [3], [52]. Understanding the rationale behind automated classification is key to fostering user trust and aligning it with broader AI ethics initiatives.

In recent years, numerous systematic reviews have addressed fake news detection from various disciplinary and methodological perspectives. However, notable gaps remain in the literature. For example, [1] proposed a comprehensive taxonomy of 17 media bias types and reviewed bias detection systems, but did not delve into algorithmic performance or real-time adaptability. Similarly, [2] focused on harmful content moderation techniques but discussed fake news only peripherally. While [52] introduced an insightful taxonomy covering feature types and model comparisons, it fell short of synthesising insights from user metadata, zero-shot learning, and multimodal integration. Technical surveys by [5] and [55] offered deep learning overviews but often treated models in isolation, omitting discussions on comparative efficiency, scalability, and cross-platform generalisability.

In contrast, our review bridges these gaps by offering a multifaceted synthesis of fake news detection research that spans technical innovation and contextual understanding. First, we evaluated and compared the performance of the traditional ML and deep learning models in terms of accuracy, efficiency, and scalability across diverse datasets. Second, we incorporated underexplored aspects, such as user metadata and contextual cues, which are often neglected in content-centric studies. Third, we highlight the potential of few-shot and zero-shot learning, which enables better adaptability in low-resource and novel misinformation scenarios. Additionally, we investigated transfer learning and multimodal architectures that integrate textual, visual, and behavioural data, emphasising their relevance to dynamic real-world applications. Through this comprehensive lens, our review consolidates prior research and proposes a forward-looking framework that prioritises robustness, real-time performance, and cross-modal generalisation.

The field is also witnessing an increase in interdisciplinary collaboration. Insights from social sciences, psychology, and media studies are being incorporated to understand the behavioural and societal dimensions of misinformation [48], [50], [57]. This convergence reflects a shift from purely algorithmic approaches to more holistic strategies for combating fake news in media. Table 1 provides a systematic overview of the previous studies, highlighting the diversity of the approaches and methodologies employed. Several key patterns emerged in this study.

First, there is a clear trend towards the adoption of more sophisticated machine learning and deep learning techniques. Although traditional methods remain relevant [3], [52], [53], the field increasingly favours deep learning models because of their superior contextual understanding.

Second, the heterogeneity of research approaches is evident. Studies range from literature [1], [2], [51] to empirical model evaluations [50], [52] and comparative analyses [49], [53]. This diversity underscores the complexity of fake news detection, and the need for comprehensive and integrative strategies. Notably, several studies [2], [51], [56], [57] have stressed the value of multimodal detection techniques that combine text with visual or network-based data. For example, [51] used deep learning techniques to detect harmful content by fusing data from multiple modalities.

Finally, Table 1 illustrates the growing interest in the social context of research on misinformation. Studies such as [1], [2], [57], and [58] explore related concerns including media bias, content moderation, and the psychological drivers of fake news sharing. For example, [5] presented an extensive evaluation of ensemble methods, graph-based techniques, and transfer-learning strategies. These studies collectively highlight the shared recognition that effective detection must address not only the content of misinformation, but also the broader ecosystem in which it spreads.

In conclusion, the field of fake news detection is evolving rapidly. Although notable strides have been made in deep learning and NLP [5], [49], [52], challenges related to scalability, adaptability, and interdisciplinary integration persist. Future research is likely to prioritise more resilient and adaptable models, with an increased emphasis on real-world applicability and interpretability. By combining advanced AI techniques with insights from social sciences and network analysis, the next generation of detection systems promises to enhance the credibility and trustworthiness of digital information ecosystems [48], [50], [57].

III. METHODOLOGY

This systematic review summarises current research and highlights the trends in the use of advanced machine learning and deep learning methods to detect fake news. Our approach involves searching for existing studies, carefully selecting relevant papers, and extracting important data on the techniques and features used. We followed the PRISMA2020 guidelines [59] to ensure a comprehensive and unbiased review. Our process included several key steps: defining the Research

TABLE 1. Previous SLR and review studies in the field of fake news detection.

Ref.	Focus	Detection Mechanisms	Features Extraction	Techniques Used	Methodology	Limitations
[48]	A comprehensive examination of deepfake-generated media and detection methodologies	Visual, Audio, and Text-based deepfake detection methods	-	NLP, ML, DL	PRISMA approach, systematic collection and analysis of 255 articles	Potential exclusion of recent articles and reliance on theoretical impacts.
[49]	Combating fake news using stance detection and subjectivity analysis	Text	-	Transformer-based language models (e.g., BERT, RoBERTa)	Fine-tuning and comparison of models	No explicit acknowledgement of limitations.
[50]	Impact of text preprocessing on TC performance	Text	Text preprocessing	Various text preprocessing Modern Transformers, traditional ML classifiers	Evaluation of preprocessing impact on classification performance	Limited preprocessing techniques and evaluation of advanced models explored.
[51]	Multimodal fake news and harmful language detection	Visual, Text	-	Multimodal data fusion methods Deep learning techniques, neural networks	Comprehensive review	Inadequate datasets and challenges in multimodal content understanding.
[52]	Taxonomy and comparative study of fake news detection techniques	Text	News content and social context	ML, DL, ensemble learning classification algorithms (e.g., SVM, Random Forest, Neural Networks)	An empirical study evaluating accuracy and computational cost	Results not representative of all fake news detection methods (lack of cross-dataset analysis).
[53]	Comparative study of ML and DL techniques for fake news detection	Text	-	ML, advanced ML, and DL transformer-based models	Benchmarking across different datasets	Insufficient data and inconsistent performance across different datasets (focused on English-only datasets).
[57]	Applications of social network analysis using deep learning	Text	-	Deep learning techniques (graph neural networks), sentiment analysis, text classification, recommender systems	Review and summary of existing research	Need for specialized hardware and potential for computational errors (limited temporal scope affects current relevance).
[1]	Media bias detection and state-of-the-art automatic detection systems	Text	-	Automatic media bias detection systems (NLP techniques, sentiment analysis, topic modelling)	Comprehensive literature review	Focus on machine learning methods and contextual limitations. The inclusion of non-English papers increases the scope.
[2]	Detection and moderation of detrimental content on social media	Visual, Text	-	AI, ML, DL content moderation algorithms hate speech detection, cyberbullying detection	Review of existing literature and research	Limited research on automated moderation and potential annotation bias.
[3]	Analyzing and detecting fake news	Text	-	Machine learning-based methods, NLP techniques	Review of methodologies	Lack of benchmark datasets and imbalanced class allocation.
[54]	Algorithms and techniques for linguistic characterization and detection of fake news	Text	NLP - vectorization, dimensionality reduction	ML, information retrieval (text preprocessing, feature extraction, ML classifiers)	Survey of methods	Limited information from news content and complex linguistic constructions.
[55]	Effectiveness of deep learning techniques for detecting online-based fake news	Text, Visual	news content, textual and social context	DL architectures (CNN, RNN, LSTM, attention mechanisms, transformer models)	Categorization of existing research, analysis of evaluation metrics	Insufficient focus on feature selection and news propagation patterns (potential biases in selected studies affect reliability).
[5]	Recent advancements in fake news	Text	-	Data mining, DL, NLP, ensemble learning,	Literature review using search engines	Challenges with data quality, computational

TABLE 1. (Continued.) Previous SLR and review studies in the field of fake news detection.

Ref.	Focus	Detection Mechanisms	Features Extraction	Techniques Used	Methodology	Limitations
[56]	detection using machine learning techniques	Text, Visual	Textual, user engagement	transfer learning, graph-based techniques	Various ML and DL models detect COVID-19 fake news effectively	requirements, and result interpretability.
	DL detects COVID-19 fake news; practical applications and future improvements explored			ML (Gradient Boost, SVM, Logistic Regression), DL (RNN, LSTM, GRU), Ensemble models		Need for multi-modal and cross-lingual approaches, potential dataset bias.
[58]	Overviews fake news detection on social networks: forms, techniques, and challenges	Text, Visual	Content, user, network, distribution, temporal-based	SVM, Decision Tree, KNN, CNN, LSTM	Comprehensive literature review	Lack of robust multilingual database, difficulty in early detection.

Questions (RQ), creating Search Terms (ST), setting the criteria for including studies, pulling out relevant information, and combining findings to draw meaningful conclusions. This structured approach helps provide a clear picture of how artificial intelligence is used to tackle the problem of false information online.

A. PLANNING AND SEARCH STRATEGY

The systematic literature review employed the Web of Science (WoS) database, which is recognised for its comprehensive coverage of high-impact research across multiple disciplines, including computer science. This database integrates five major digital repositories, IEEE, Springer, Elsevier, MDPI, and ACM, as shown in Screen Figures 1, selected for their scientific rigor and extensive cross-disciplinary publications. A well-structured search query was created to optimise the retrieval of pertinent studies. The query utilised key terms such as “fake news”, “false information”, “natural language processing”, “machine learning”, and “deep learning” combined with Boolean operators to refine the results as follows:

B. SEARCH TERM

Key terms were carefully selected to construct a search string to identify studies that addressed the research question. This string incorporates synonyms and acronyms, with terms combined using the Boolean operator. The resulting search string was as follows: (“fake news” OR “false information”) AND (“natural language processing” OR “NLP” OR “text analysis” OR “text classification” OR “sentiment analysis”) AND (“machine learning” OR “deep learning”). The search was restricted to publications from 2020 to 2024 to ensure the inclusion of recent articles and focused on keywords, titles, and abstracts to maintain relevance. Only full articles and review papers were included to maintain high quality standards.

The final search, conducted on 17 July 2024 aimed to identify the most relevant and current studies on advanced computational techniques for detecting false information on social media. This systematic review adhered strictly to the

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, ensuring comprehensive and transparent reporting. The paper selection process, in alignment with the PRISMA methodology, is visually represented in a flow diagram (Figure 4) to ensure transparency and reproducibility. This approach facilitated a thorough and systematic exploration of the literature related to machine- and deep-learning approaches for fake news detection, recognising the interdisciplinary nature of the field.

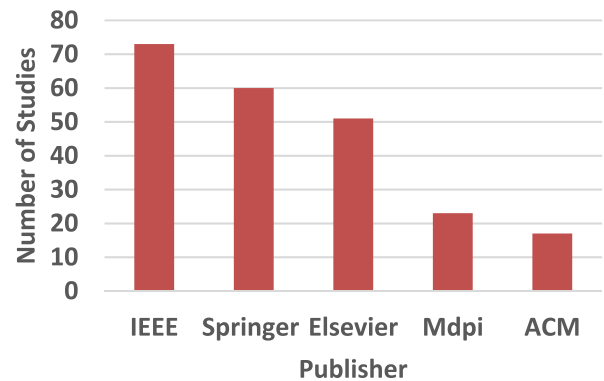


FIGURE 1. Number of papers based on publisher.

This study bridges the domains of fake news detection, social media, natural language processing, and machine learning, thereby providing a robust analysis of advanced computational techniques for identifying false information. A PRISMA flow diagram was used to visually represent the study selection process. This diagram illustrates the number of records identified through database searches based on the selected criteria. The diagram details the reasons for exclusion at each stage, thus enhancing the transparency and reproducibility of the study selection process.

C. SELECTION CRITERIA AND SCREENING PROCESS

To maintain the rigor of this review, explicit criteria for the inclusion and exclusion of papers were established as follows. These criteria were pivotal in navigating the extensive literature, ensuring the selection of studies that were not only

germane to the research question but also met rigorous academic standards. The systematic selection approach helped maintain the relevance and quality of the included studies.

1) INCLUSION CRITERIA

The inclusion criteria were carefully designed to capture the most relevant and recent research in the field, as shown in Table 2. These criteria ensured that the selected papers were not only pertinent to the research topic but also reflected the latest developments in the rapidly evolving field of fake news detection. The temporal scope of 2020–2024 was selected due to the sharp increase in misinformation during the COVID-19 pandemic. According to World Health Organization (WHO), Pan American Health Organization (PAHO), and United Nations Educational, Scientific and Cultural Organization (UNESCO) reports, the global health crisis triggered an unprecedented wave of online misinformation, termed an “infodemic”, characterised by millions of posts spreading false or misleading claims [60], [61], [62]. Studies have also demonstrated a substantial increase in research efforts related to fake news detection and the development of detection systems during this period [63], [64]. Following the initial screening phases, a fourth criterion was implemented to further align the selected studies with the defined research questions. This additional step ensured that the final set of papers directly addressed the research objectives and provided highly relevant insights into the challenges and advancements in fake news detection.

TABLE 2. Inclusion criteria.

Inclusion criteria
Papers with a primary focus on fake news or related terms.
English language publications.
Publications from the timeframe of 2020 to 2024.
Research area of Computer Science.
Papers that answered one or more of the research questions fully or practically.
Papers classified as Articles or Review papers.

2) EXCLUSION CRITERIA

The following exclusion criteria were adopted to refine the selection process, as shown in Table 3. These criteria helped filter out less relevant or potentially problematic studies, ensuring that the review remained targeted and manageable.

TABLE 3. Exclusion criteria.

Exclusion criteria
Papers are not primarily focused on fake news.
Non-English language papers.
Papers that do not directly answer the research questions.
Papers without abstracts.
Inaccessible papers.

3) SCREENING AND SELECTION PROCESS

The authors employed a systematic screening process, beginning with an initial database search that yielded 406 papers from the Web of Science. Through a series of well-defined filtering stages guided by established inclusion and exclusion criteria, the selection was refined to a final set of 90 papers. Each stage was designed to ensure that the included studies were relevant and of high methodological and thematic quality. Initially, applying the publication period filter (2020–2024) reduced the number of papers to 356, all of which were in English. This timeframe was strategically selected to capture the most recent developments in fake news detection, particularly during and after the COVID-19 infodemic, as illustrated in Figure 2. Further refinement limited the scope to the field of Computer Science, narrowing the list to 278 papers and ensuring technical relevance.

Subsequently, the authors focused on papers from major, reputable academic publishers, yielding a dataset of 224 papers distributed as follows: IEEE (73), Springer (60), Elsevier (51), MDPI (23), and ACM (17). This ensured that the review included only peer-reviewed publications recognised for their academic rigor. Among these, 132 were classified as research articles and 11 as review papers, both of which typically offer substantial scholarly contributions.

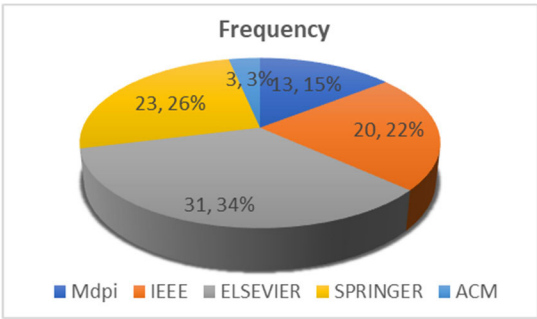


FIGURE 2. Number of papers by publisher.

The final screening phase applied stricter criteria to ensure alignment with the study objectives. A total of 36 papers were excluded for not directly addressing any of the defined research questions, and 17 were excluded for relying on non-English datasets, which could limit the consistency of the analysis and introduce language-related biases. One paper was removed due to the absence of an abstract, which is a key element for the initial relevance assessment.

To ensure consistency during the screening process, each article was independently reviewed by two authors using a structured evaluation rubric aligned with the research questions (Q1–Q5). Discrepancies in the inclusion decisions were resolved through discussion and consensus. Although no formal inter-rater reliability metric was calculated, the use of a standardised scoring matrix and collaborative review ensured

transparency, minimised bias, and supported consistent decision making.

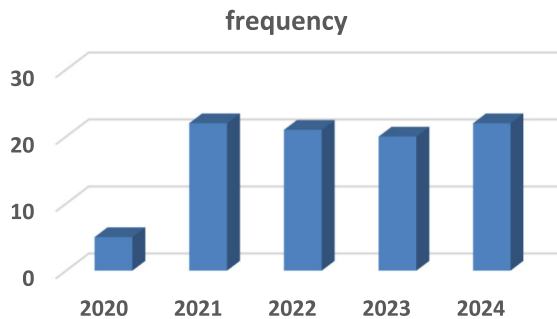


FIGURE 3. Number of published papers selected by year.

Ultimately, as shown in Figure 3, 90 papers met all the criteria and were selected for the full analysis. This multistage, transparent selection process provides a solid foundation for the review, ensuring that it is based on the most recent, relevant, and methodologically sound research in the field of fake news detection using machine and deep learning approaches. The rigor of this process enhances the credibility and reproducibility of these findings.

4) RESEARCH OBJECTIVE AND QUESTIONS

This systematic literature review (SLR) explores recent advances in text-based fake news detection on social media, focusing on machine learning and deep learning approaches. This study adopted strategies from [1], [65], [66], and [67] to conduct a comprehensive review. Five research questions were formulated to guide the investigation, addressing the crucial aspects of fake news detection. These questions encompass the incorporation of user metadata and social network structure (Q1), the design of interpretable machine learning models (Q2), the comparison of deep learning models with traditional machine learning algorithms (Q3), the potential of pre-trained language models (Q4), and the performance of few-shot and zero-shot learning techniques (Q5). Table 4 illustrates the citations of articles based on their alignment with the research questions.

Figure 5 illustrates the distribution of studies addressing the five research questions (RQs) within the review. The most frequently investigated area was RQ3, which focused on the comparative performance of deep learning models versus traditional machine learning algorithms in fake news detection. This was followed by RQ4, which explored the role of pre-trained language models in enhancing the performance of fake news detection systems. While RQ1 and RQ2, respectively, concerning the impact of user metadata and interpretability of models, garnered moderate attention, RQ5, addressing few-shot and zero-shot learning for novel fake news detection, received the least focus. This distribution suggests a strong emphasis on foundational model comparison and language model integration in the field, with

emerging areas such as interpretability and few-shot learning warranting further exploration.

5) QUALITY ASSESSMENT

In addition to the established inclusion and exclusion criteria, each selected article underwent a structured quality assessment to ensure the integrity and rigor of this review. This evaluation was essential for mitigating potential bias and reinforcing the reliability and credibility of the findings. Aligned with the methodological standards of a systematic literature review (SLR), the quality assessment process ensured that only methodologically sound and thematically relevant studies were included in the review.

To guide this process, each study was evaluated against several criteria: clarity of research objectives, appropriateness of the study design or methodology, dataset characteristics (including size, language, and class balance), transparency and reproducibility of experimental procedures, and completeness of reported performance metrics (such as accuracy, precision, recall, and F1-score). These indicators ensured consistency and depth in evaluating the methodological robustness of the included studies.

A key component of this assessment was evaluating how each study directly or partially addressed one or more of the predefined research questions (RQs). A study was considered to partially address an RQ if it contributed meaningfully to at least one of them, for example, by proposing detection techniques, analysing evaluation metrics, or identifying specific challenges, even if it did not address the entire research scope.

Each article was independently reviewed by the two authors using a structured evaluation rubric aligned with the five research questions (Q1–Q5). Discrepancies in the inclusion decisions were resolved through discussion and consensus. Although no formal inter-rater reliability metric (such as Cohen's Kappa) was calculated, the use of a standardised scoring matrix and collaborative review process ensured transparency, consistency, and minimisation of bias.

This rigorous approach, when combined with well-defined inclusion and exclusion criteria, provides a strong methodological foundation for this review. The focus on recent research (2020–2024), in the context of significant developments influenced by the COVID-19 pandemic, further enhances the findings' timeliness and relevance. Moreover, the structured quality assessment contributes to the transparency and reproducibility of the review, both of which are essential for high-impact academic research. Table 5 illustrates the criteria used to assess the methodological rigor and thematic relevance of the included studies during the quality assessment phase.

IV. ANALYSIS AND DISCUSSION

The field of fake news detection has witnessed significant advancements in recent years, particularly in feature extraction techniques and model architectures. There is a clear trend towards more sophisticated approaches, with deep learning models, especially transformer-based architecture

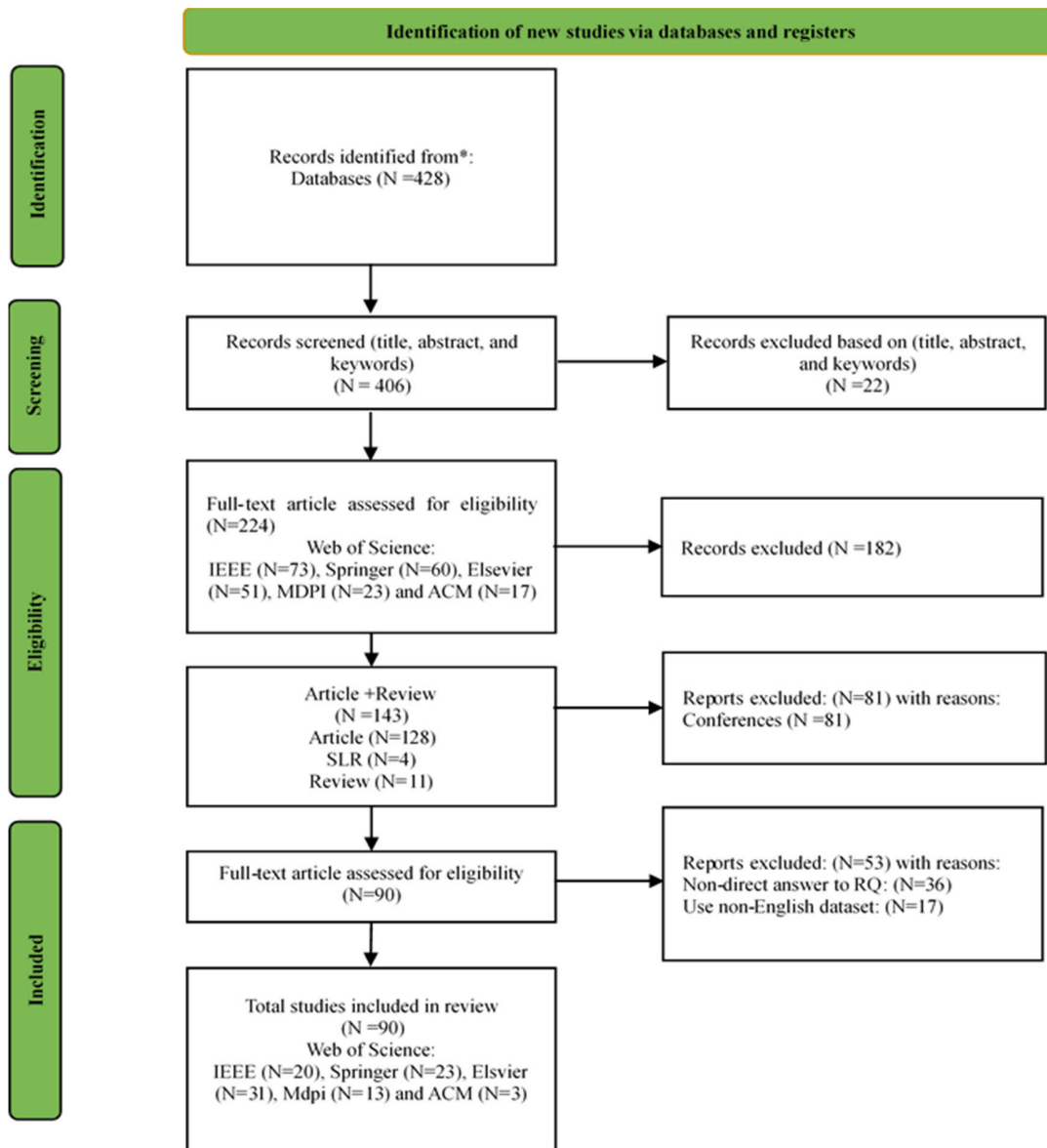


FIGURE 4. Prisma flow diagram.

such as BERT, demonstrating superior performance compared to traditional machine learning methods. For instance, [47] achieved an impressive 99.96% accuracy using BERT, substantially outperforming previous approaches. The integration of linguistic features, user behaviour, and social network information is crucial for comprehensive fake news detection systems. Researchers are increasingly focusing on multimodal approaches that combine textual, visual, and social context data to enhance the accuracy of detection. Transfer learning and cross-lingual models have shown promise in addressing the challenges of limited labelled data in new domains or languages. However, as models become more complex, there is a growing emphasis on developing interpretable systems, with studies such as [25] and [44] exploring techniques to make deep learning models more transparent.

Few-shot and zero-shot learning approaches have emerged as promising solutions, offering notable advantages over traditional machine learning methods in addressing the dynamic landscape of misinformation. These techniques are particularly relevant for fake news detection, as they help overcome critical challenges, such as rapidly evolving misinformation topics and the scarcity of labelled data. Few-shot learning typically relies on meta-learning algorithms or fine-tuning of pre-trained models (e.g. BERT, RoBERTa) with a limited number of labelled examples, enabling models to generalise with minimal supervision [18].

In contrast, zero-shot learning often leverages prompt-based frameworks or semantic similarity in embedding spaces to classify instances from completely unseen classes or domains. These methods offer several practical advantages. First, they demonstrate an enhanced ability to detect

TABLE 4. Frequency of articles aligned with the research questions.

No.	Research Questions	References
1	How does incorporating user metadata and social network structure improve the performance of machine learning models in identifying fake news compared to text-only approaches?	[6], [12], [14], [16], [17], [21], [24], [26], [28], [31], [32], [34], [39], [40], [41], [43], [44], [58], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79]
2	How can we design machine learning models for fake news detection that provide interpretable results without significantly sacrificing performance?	[13], [25], [27], [33], [35], [42], [44], [68], [77], [79], [80], [81], [82], [83]
3	How do deep learning models perform compared to traditional machine learning algorithms in terms of accuracy, efficiency, and generalizability for fake news detection?	[7], [8], [9], [10], [12], [14], [15], [16], [17], [19], [20], [21], [23], [24], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [39], [40], [42], [43], [44], [45], [46], [47], [56], [58], [68], [71], [72], [73], [74], [75], [76], [77], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102]
4	To what extent can pre-trained language models enhance the performance and adaptability of fake news detection systems across various topics and languages?	[7], [8], [12], [13], [14], [15], [17], [18], [19], [24], [27], [30], [31], [34], [37], [39], [40], [43], [45], [53], [58], [70], [71], [75], [76], [78], [81], [82], [83], [84], [86], [88], [90], [91], [94], [99], [100], [101], [102]
5	How well do few-shot and zero-shot learning techniques perform in identifying novel types of fake news compared to traditional supervised learning methods?	[18], [47], [86], [95], [100], [103]

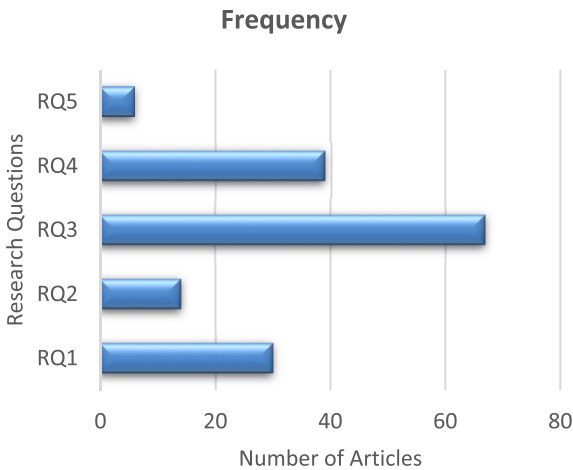


FIGURE 5. Frequency of articles based on alignment with the research questions.

emerging forms of fake news with minimal additional training data, helping to address the “cold start” problem more effectively than traditional models that require extensive retraining [86], [100], [103]. Second, they significantly reduce the dependence on large labelled datasets, which are costly and time-consuming to compile, especially in high-stakes domains such as health or politics [103]. Additionally, some implementations may offer improved

TABLE 5. Criteria used to assess the methodological rigor and thematic relevance of included studies during the quality assessment phase.

Criterion	Description
Clarity of Research Objectives	The study clearly defines its goals, research questions, or hypotheses.
Appropriateness of Study Design	The chosen methodology aligns well with the research goals and is logically structured.
Dataset Characteristics	The dataset is well-described, with details on size, language, class balance, and source.
Transparency and Reproducibility	Experimental setup, tools, models, and data are adequately explained to allow reproducibility.
Performance Reporting	The study reports standard evaluation metrics (e.g., accuracy, precision, recall, F1-score).
Relevance to Research Questions (RQs)	The study fully or partially addresses one or more of the review’s predefined research questions.

interpretability by highlighting the linguistic or contextual reasoning underlying their classifications.

However, despite their potential, these approaches have important limitations. Domain sensitivity, prompt instability, and inconsistent performance across nuanced misinformation categories remain active challenges to be addressed. For example, [86] applied a few-shot fine-tuning approach using BERT but observed performance degradation when tested on entirely new misinformation topics, highlighting the need

for more robust generalisation strategies. Therefore, ongoing research is essential to enhance the stability, transferability, and interpretability of these models before they can be reliably deployed in real-world fake news detection systems.

Notably, among the reviewed studies, few-shot and zero-shot learning (RQ5) appeared the least frequently. This reflects a broader research trend rather than a selection bias, as these techniques are still in the early stages of adoption. Their limited representation underscores a valuable opportunity for future research, particularly in addressing low-resource scenarios and improving model adaptability in evolving misinformation environments.

A. CLASSIFICATION OF FAKE NEWS DETECTION APPROACHES

In the evolving landscape of fake news detection, researchers have employed a diverse array of methodologies that can be broadly classified into three main approaches: Deep Learning (DL), Traditional Machine Learning (ML), and Natural Language Processing (NLP). These classifications, while distinct, often overlap and complement each other in modern fake news detection systems, leading to the development of hybrid approaches that leverage the strengths of multiple techniques to achieve better performance. Figure 6 provides a visual overview of the major methods applied in fake news detection, highlighting key models and their interconnections.

1) DEEP LEARNING APPROACHES

As illustrated in Figure 6 Deep Learning (DL) approaches have gained significant traction in recent years owing to their ability to automatically learn complex features from raw text data. Among the foundational models, Convolutional Neural Networks (CNN) [7], [76], [96], [97] are effective at capturing local patterns in text sequences, while Recurrent Neural Networks (RNN) [7], [55], [76], [97], including their variants such as LSTM and Bi-LSTM, have demonstrated remarkable success in capturing spatial and temporal patterns in text making them well-suited for modelling sequential dependencies and long-term context. More advanced architectures, such as Hierarchical Attention Networks (HAN) [30], [56] provide structured representations by attending to informative words and sentences within documents. More recent breakthroughs stem from transformer-based models, including BERT and RoBERTa [27], [34], [45], [86], [101], which excel at capturing bidirectional context and semantic nuances. Other promising DL models including Graph Convolutional Networks (GCNs) [43], which capture relationships across documents nodes, and Deep Autoencoders [24], which are used for unsupervised feature learning and dimensionality reduction.

2) MACHINE LEARNING METHODS

Machine Learning methods continue to play a crucial role in fake news detection, especially in scenarios where labelled data is limited, model interpretability is essential, or

computational resources are constrained. As shown in Figure 6, traditional ML algorithms are often used either as standalone classifiers or in combination with feature engineering and ensemble techniques to enhance prediction accuracy. One of the most widely used techniques is the Support Vector Machines (SVM) [7], [82], [90], known for its effectiveness in high-dimensional spaces and its robustness in handling linearly and non-linearly separable data. Similarly, Random Forests [7], [82], [87], are popular due to their ensemble structure, which combines multiple decision trees to reduce overfitting and improve generalization. Logistic Regression (LR) [82], [90] remains a baseline model in many studies for its simplicity, takes when paired with strong feature representations. While, Naïve Bayes classifiers [82], [90], [91], particularly Multinomial and Bernoulli variants, are favoured for their probabilistic foundation and efficiency in large-scale text classification problem. Likewise, Decision Trees [82], [90] provide a visual and interpretable model structure that is easy to analyse, although they are prone to overfitting if not combined with pruning or ensemble methods. k-Nearest Neighbors (k-NN) [82], [90], is another technique employed in fake news detection, especially when simplicity and instance-based learning are desirable. While it is computationally expensive during inference, it can be highly effective with well-engineered feature spaces. Finally, ensemble learning has emerged as a powerful strategy to improve model robustness and accuracy. Methods such as AdaBoost and XGBoost [7], [29] combine multiple weak learners into a strong predictive model, often outperforming individual classifiers. These ensemble approaches are particularly valuable when dealing with imbalanced datasets or heterogeneous feature space.

3) NATURAL LANGUAGE PROCESSING

Natural Language Processing (NLP) techniques form the foundation for text preprocessing and feature extraction in many fake news detection systems as shown in Figure 6. From basic methods such as Bag-of-Words (BoW) [24], [54] and Term Frequency-Inverse Document Frequency (TF-IDF) [90], [98] to more sophisticated techniques such as Word Embeddings such as Word2Vec and GloVe [47], [90], [96], which capture contextual meaning. N-grams [98] help model local word sequences, while Named Entity Recognition (NER) [77] extracts proper names and entities relevant to misinformation. Sentiment analysis [12], [39] and topic modelling [12] are applied to assess opinion polarity and thematic structure. Additionally, linguistic feature extraction [77], [82] supports model interpretability and rule-based filtering. NLP approaches enable the extraction of meaningful features from text data. These features can then be utilised by both traditional ML algorithms and deep learning models to improve classification accuracy.

4) HYBRID APPROACHES

The most promising advancements in fake news detection often arise from hybrid approaches that combine elements

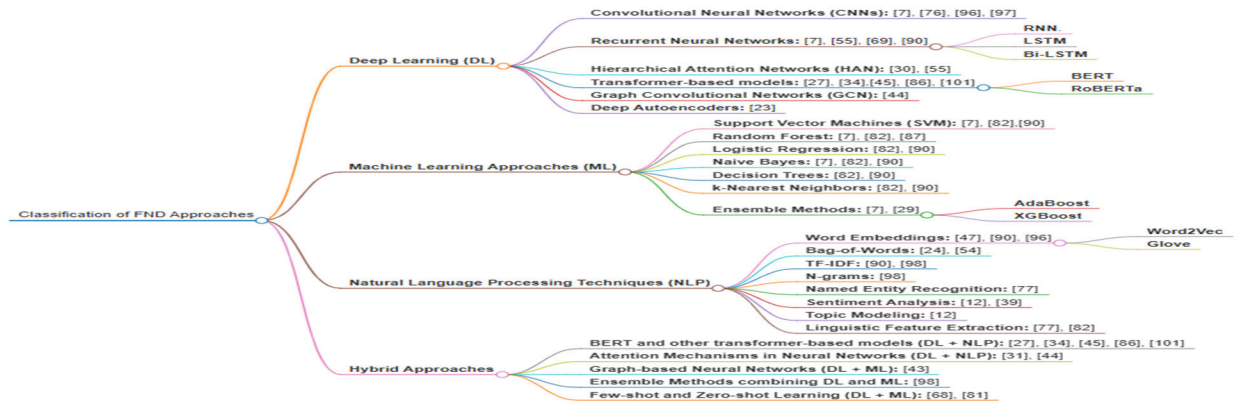


FIGURE 6. Classification of Fake News Detection (FND) Approaches. This mind map illustrates the main categories of approaches used for fake news detection: Deep Learning (DL), Machine Learning (ML), Natural Language Processing (NLP), and Hybrid methods. Each branch details the specific models or techniques applied under that category, with relevant references. Hybrid approaches combine elements from DL, ML, and NLP.

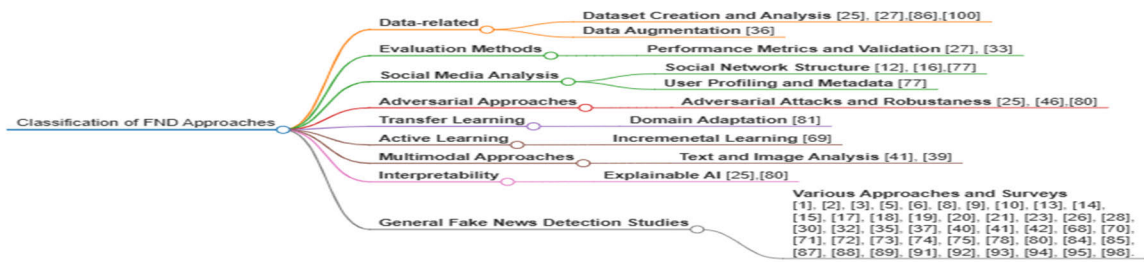


FIGURE 7. Expanded Research Landscape in Fake News Detection (FND). This mind map presents additional dimensions that enrich the FND research landscape beyond core classification approaches.

from several categories as presented in Figure 6. For instance, transformer-based models, such as BERT [27], [34], [45], [86], [101], inherently integrate deep learning architectures with sophisticated NLP principles. Similarly, attention mechanisms in neural networks [31], [44] leverage both DL and NLP concepts to enhance the model performance. Graph-based Neural Networks [43] combine elements of DL and ML to model complex relationships in data, whereas ensemble methods [98] that integrate DL and ML algorithms aim to leverage the strengths of both approaches. The choice between these approaches is often influenced by factors such as dataset size and characteristics, available computational resources, the need for model interpretability, and specific requirements of the fake news detection task. For instance, in scenarios with limited labelled data, few-shot and zero-shot learning techniques [86], [100], [103] that combine DL and ML principles may be more suitable. Conversely, when dealing with large-scale datasets and complex linguistic patterns, deep learning models may offer superior performance.

The field of fake news detection encompasses not just machine learning and NLP techniques, but also various other aspects such as data collection, evaluation methodologies, and robustness against adversarial attacks. To give a more complete picture of the diverse areas of study within the field, we would need to expand our analysis beyond just the ML,

DL, and NLP approaches to include these broader aspects of fake news detection research. In addition, Figure 7, illustrated several auxiliary and emerging strategies contribute significantly to fake news detection beyond the core classification methods. These include data-related approaches such as dataset creation [25], [27], [86], [100], and data augmentation [36], which improve model robustness. Evaluation methods [27], [33] ensure accurate performance measurement. Social media analysis explores network structure and user metadata [12], [16], [77], while adversarial approaches address vulnerabilities through robustness testing [25], [46], [80]. Transfer learning [81], active learning [69], and multimodal approaches [39], [41], support learning efficiency and handle multi-source inputs. Interpretability techniques such as Explainable AI [25], [80] aim to enhance model transparency. Collectively, these approaches enrich the FND landscape by addressing challenges in data, evaluation, robustness, and generalisability.

B. SYNTHESIS OF FINDINGS

To conduct a comprehensive analysis and comparison of the approaches used for fake news detection, we systematically examined the methodologies, techniques, and outcomes reported in the selected literature, with a particular focus on how these approaches address the specific research questions

outlined in our Systematic Literature Review (SLR). This analysis involved a detailed evaluation of various machine and deep learning models, feature extraction techniques, and dataset characteristics employed in each study. We will assess the effectiveness of different approaches in terms of their accuracy, scalability, and ability to handle diverse types of fake news on various platforms. Additionally, we explore how these methods tackle challenges such as the evolving nature of misinformation, multimodal content, and the need for explainable AI in fake news detection. By aligning our analysis with the SLR research questions, we aimed to identify patterns, trends, and gaps in the current research, ultimately providing insights into the most promising directions for future work in the field of automated fake news detection.

1) OVERVIEW OF THE MODELS' PERFORMANCE

Recent studies on fake news detection have employed various approaches, with a notable trend towards deep learning models. Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, Hierarchical Attention Networks (HAN), and transformer-based models such as BERT have been widely used in this domain [7], [27], [31], [34], [44], [45], [55], [76], [86], [96], [97], [101]. These deep learning models have generally demonstrated superior performance compared to traditional machine learning algorithms, such as Support Vector Machines (SVM), Random Forest, Logistic Regression, and Naive Bayes [7], [82], [90]. The strength of deep learning models lies in their ability to handle high-dimensional data, capture complex patterns, and learn hierarchical representations of text [31], [44], [55], [76]. CNNs, LSTMs, and their hybrid variants have demonstrated remarkable accuracy, often exceeding 90% in various studies [96], [97], [101]. For instance, the FNDNet model achieved an impressive 98.36% accuracy [96], whereas another CNN model achieved 94% accuracy, specifically for COVID-19-related misinformation [97], as presented in Table 7.

However, these advantages are associated with increased computational requirements and reduced interpretability compared to traditional models [44], [47]. As the field progresses, there is a need to balance performance gains with practical deployment considerations, potentially through a dual approach of scaling up models while developing more efficient and specialised architecture for resource-constrained environments.

2) PRE-TRAINED LANGUAGE MODELS AND TRANSFER LEARNING

Pre-trained language models, such as BERT, RoBERTa, and ELMo, have emerged as powerful tools for enhancing fake news detection systems. These models excel at capturing contextual information across various topics and languages, enabling a deeper understanding of the subtle complexities often present in misinformation [27], [34], [86], [100]. Their versatility allows models trained on general news to perform well, even in specialised domains, addressing the challenge

of misinformation across multiple fields [34]. Transfer learning, particularly fine-tuning pre-trained language models, has led to state-of-the-art performance, even with limited labelled data in the target domains [86], [100]. This approach has shown particular promise in addressing the challenges of fake news detection in low-resource language scenarios. More importantly, pre-trained models can be fine-tuned on smaller, task-specific datasets while maintaining high performance, mitigating the scarcity of large-scale, up-to-date labelled data in the rapidly evolving landscape of fake news [101]. However, these advantages are accompanied by notable challenges. Computational complexity can limit real-time detection or deployment on resource-constrained devices [45]. Additionally, models may inherit biases from their pre-training data, which affect their performance in diverse cultural contexts [100]. Their “black box” nature also poses issues for explainability, which is a crucial factor in building trust and understanding model weaknesses in fake news detection [101]. To address these challenges, researchers must carefully consider fine-tuning strategies, model selection, and potential ensemble approaches that combine pre-trained models with other techniques [27], [34], [100], [101].

3) MULTIMODAL APPROACHES AND SOCIAL CONTEXT INTEGRATION

To improve the accuracy and robustness of fake news detection, researchers have explored the integration of user metadata with social network information. Techniques such as Graph Convolutional Networks (GCN), user profiling, and social network analysis have been employed to consider the propagation patterns of fake news and identify influential users [43], [77]. Several studies have demonstrated that combining content-based features with user profile information, propagation patterns, and network structures can lead to substantial improvements in detection accuracy [26], [34], [43], [44]. For example, the Graph Global Attention Network with Memory (GANM) approach [43] integrates two complementary modules: a Global Attention Module with Memory that captures structural patterns across multiple graphs using a GRU-based memory mechanism, and a Partial Key Message Learning module that highlights essential node-level and graph-level features. This dual mechanism allows the GANM to model both global and local propagation structures effectively, achieving higher classification performance than existing baselines on real-world datasets such as Politifact, Gossipcop, and Weibo.

Beyond the social context, multimodal integration, which combines text, images, and auxiliary metadata, has demonstrated significant performance improvements. In [76], a multimodal CNN incorporating textual and visual features achieved 87% accuracy on the Fakeddit dataset, outperforming a text-only BERT model, which achieved 78%, thereby realising a 9% improvement. Similarly, [39] introduced an ensemble learning framework (ELD-FN) that combines textual sentiment and visual features using V-BERT. This

model surpassed the baseline systems with improvements of 12.57%, 18.15 %, and 12.58% in accuracy, recall, and F1-score, respectively. The SSM framework proposed in [41] further advanced multimodal robustness by integrating stylistometric and semantic similarity features and consistently outperformed the unimodal baselines across three benchmark datasets. Additionally, [14] addressed “cheapfake” detection using a transformer-based text-image matching model, which achieved superior performance on the COSMOS dataset compared to unimodal baselines, demonstrating the effectiveness of aligning textual semantics with visual context. These findings affirm that multimodal approaches not only enhance classification performance but also improve model resilience in noisy real-world settings, particularly in detecting nuanced misinformation, such as satire, manipulated content, and false connections. This multimodal approach, which considers both the content and context of information spread, aligns with social science theories regarding the role of echo chambers and information cascades in misinformation diffusion. However, the integration of user metadata and social network signals raises important ethical considerations, particularly regarding privacy. Models that leverage behavioural signals, profile attributes, and online interaction patterns may inadvertently expose sensitive user information or operate without adequate consent. Thus, future research should consider privacy-preserving techniques and responsible data governance frameworks when designing socially informed fake news detection systems [43], [77]. While these approaches have shown considerable promise in enhancing detection performance, they also introduce additional complexity to model design and computation, as well as ethical challenges related to user data.

4) INTERPRETABILITY AND EXPLAINABLE AI

As fake news detection systems become increasingly sophisticated, the demand for interpretability has become critical, particularly in high-stakes domains such as health, politics, and public safety. To enhance trust and transparency, recent research has focused on explainable AI (xAI) techniques that provide insights into the decision-making processes of models.

Post-hoc explanation tools, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), have been widely adopted to identify the contribution of individual features to predictions. For example, [27] demonstrated the use of SHAP to explain misinformation classifications during the COVID-19 infodemic, helping users understand why specific claims were highlighted. Reference [33] developed an interactive browser-based tool that visualised news credibility assessments, improving user comprehension and confidence in model decisions.

Beyond post-hoc methods, integrated explainability has shown promise. Reference [44] introduced a pyramidal co-attention network that jointly processes article content and user comments, highlighting the key textual components

that justify fake news predictions. This approach not only improves the classification performance but also offers interpretable rationales, enabling end users to trace the model’s reasoning.

However, the implementation of xAI methods introduces new challenges. Reference [25] raised concerns about the security of explanation techniques, showing that SHAP-based outputs can be adversarially manipulated to mislead both models and users. This reveals a crucial trade-off between transparency and robustness, underscoring the need for secure and trustworthy explanation mechanisms.

Among the 90 studies reviewed, approximately 18 (20%) explicitly incorporated interpretability techniques into their fake news detection models. These approaches primarily involve integrated attention mechanisms, co-attention networks, post-hoc explanation tools such as SHAP and LIME, and the visualisation of key feature contributions. Despite this growing interest in XAI, only a limited number of studies have conducted user-centred evaluations or applied formal interpretability metrics to assess the quality or impact of these explanations. This gap highlights the need for the systematic adoption and evaluation of interpretability frameworks in future fake news detection research, particularly to enhance transparency, user trust, and ethical accountability in real-world applications.

Overall, many deep learning models for fake news detection remain opaque, with limited consideration of how explanations are perceived or utilised by end users. A persistent trade-off exists between maximising the predictive performance and ensuring interpretability. Future research should prioritise the development of explainable systems that balance these objectives and address adversarial vulnerabilities. Human-centred evaluation frameworks are essential for assessing the usability, effectiveness, and psychological impact of explanations in practical misinformation detection scenarios.

5) ADAPTABILITY AND EMERGING TECHNIQUES

The rapid evolution of fake news necessitates the development of models that can adapt to novel types with limited labelled datasets. Few-shot and zero-shot learning approaches, including transfer and active learning techniques, have shown potential in this regard [62], [74]. Techniques such as topological data analysis (TDA) have been shown to significantly improve the accuracy of deep learning models when trained on very small datasets (50–400 observations) [45]. Additionally, ensemble learning and hybrid approaches have been effective in addressing the challenges of limited data and domain shifts in fake news detection [88], [91]. These advancements are particularly important given the rapidly evolving nature of misinformation tactics and the need for agile responses to emerging issues. Researchers have stressed the importance of adaptability in detection approaches. This includes continually updating datasets and models to reflect new trends in misinformation [10], [88], and exploring semi-supervised and unsupervised learning

techniques to address the ongoing challenge of limited labelled data [24], [77].

Table 6 presents a comprehensive summary of the methodologies employed in fake news detection, aligned with our research questions (3.3.4). This synthesis enables a clear visualisation of the comparative strengths and weaknesses of different approaches. For example, while transformer-based models offer very high accuracy (90-99%) and excellent adaptability to new domains, they score lower on efficiency and interpretability. Conversely, traditional ML approaches provide high interpretability and efficiency but demonstrate only moderate accuracy (70-85%) and limited adaptability.

Notably, the expanded analysis reveals additional insights: NLP Feature Engineering methods deliver efficient processing with high interpretability but struggle with adaptability, whereas Linguistic Analysis approaches offer very high interpretability and domain adaptability despite moderate accuracy. Multimodal approaches that combine text and image analyses achieve high accuracy and generalisability but at the cost of computational efficiency.

Social Network Analysis presents a balanced profile, with particularly strong performance in domain adaptability. Furthermore, this meta-analysis of performance metrics across all examined studies provides comprehensive quantitative insights into the effectiveness of various detection methods, highlighting the inherent trade-offs that researchers must navigate when selecting appropriate techniques for fake news detection.

Where:

- *Accuracy* denotes the proportion of correct predictions made by a fake news detection model out of the total number of cases evaluated. It measures how well a model distinguishes between genuine and fake news [41]. It was calculated using the following equation:

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

where: TP: True Positives, TN: True Negatives, FP: False Positives and FN: False Negatives

- *Efficiency* refers to the ability of a model to achieve accurate results with minimal computational resource (i.e. time and memory). It is often evaluated based on the trade-off between the computational resources used and the performance. Possible metrics include training time, inference time, and memory usage [34], [83].
- *Generalisability* refers to the ability of a model to perform well on new, unseen data. It is often assessed using metrics such as accuracy, precision, recall, and F1-score, which are used to evaluate the model performance on test data [3], [33].

Generalisability

$$= |\text{Training Accuracy} - \text{Testing Accuracy}|$$

Lower values indicate better generalisability.

- *Interpretability* refers to the ease with which the predictions of a model can be explained and understood by

humans. It is often evaluated based on the model's ability to provide explanations for its predictions [23]. Possible metrics include the number of features used, depth of decision trees, and attention weights in neural networks.

- *Adaptability to New Domains* refers to a model's ability to perform well when applied to data from domains or topics different from its training data, often with minimal fine-tuning. It is often evaluated based on the model's performance on new, unseen data from different domains [13].

$$\text{Adaptability} = \frac{(\text{Performance on New Domain})}{(\text{Performance on Original Domain})}$$

A comparison of fake news detection methodologies reveals significant trade-offs among various approaches. Transformer-based models and deep learning techniques consistently outperform traditional machine learning methods in terms of accuracy, generalisability, and adaptability to novel domains. However, this superior performance comes at the cost of reduced efficiency and interpretability of the model. Traditional machine learning approaches, while less accurate and adaptable, offer advantages in terms of computational efficiency and result interpretability. Ensemble methods strike a balance by combining multiple models to achieve high levels of accuracy. The choice of methodology for fake news detection ultimately depends on specific task requirements, such as the available computational resources, the need for result interpretation, and the frequency of encountering novel types of misinformation. For many real-world applications, a hybrid approach that leverages the strengths of multiple methods may provide an optimal balance between accuracy, efficiency, interpretability, and adaptability. This synthesis of approaches could potentially address the multifaceted challenges posed by the ever-evolving fake news landscape.

C. FEATURE EXTRACTION APPROACHES FOR FAKE NEWS DETECTION

Feature extraction is a critical component of the fake news detection pipeline, serving as a bridge between raw data and machine learning models. This process involves transforming unstructured text data into structured and meaningful representations that algorithms can effectively process. The evolution of feature extraction techniques for fake news detection mirrors broader advancements in natural language processing and machine learning. Early approaches relied heavily on hand-crafted features, such as linguistic characteristics, sentiment analysis, and basic statistical measures [7], [69]. While interpretable, these methods often struggle to capture the nuanced and context-dependent nature of fake news. The advent of deep learning has revolutionised feature extraction, enabling models to automatically learn complex hierarchical representations from raw text data [96], [101]. Techniques such as word embeddings (e.g. Word2Vec, GloVe), and more recently, contextual embedding from transformer models (e.g. BERT, RoBERTa),

TABLE 6. A comparative table of the strengths and weakness of different approaches used in FND.

Approach	Accuracy	Efficiency	Generalizability	Interpretability	Adaptability to New Domains
Traditional ML (e.g., SVM, Random Forest)	Moderate (70-85%) [7], [82], [90]	High [82], [90]	Moderate [7], [90]	High [44]	Low to Moderate [81]
Deep Learning (e.g., CNN, RNN)	High (85-95%) [7], [55], [76]	Moderate [47]	High [55], [76]	Low [44]	Moderate [81]
Transformer-based (e.g., BERT, RoBERTa)	Very High (90-99%) [27], [34], [86]	Low to Moderate [45]	Very High [86], [100]	Low to Moderate [25]	High [100]
Graph-based (e.g., GCN, GANM)	High (85-95%) [43]	Moderate [43]	High [43]	Low [43]	Moderate [43]
Ensemble Methods	High (85-95%) [98]	Moderate [98]	High [98]	Moderate [98]	Moderate [98]
Interpretable Models (e.g., Attention)	High (80-90%) [31], [44]	Moderate [44]	Moderate to High [31]	Very High [44]	Moderate [31]
Few-shot/Zero-shot Learning	Moderate (70-85%) [86], [103]	High [103]	Moderate to High [100]	Moderate [103]	Very High [100], [103]
NLP Feature Engineering (e.g., TF-IDF, Bag-of-Words)	Moderate to High (75-90%) [10], [36], [77]	High [36]	Moderate [10]	High [77]	Low [36]
Linguistic Analysis	Moderate (70-85%) [77], [82]	High [77]	Moderate [82]	Very High [77]	Very High [77]
Multimodal (Text+Image)	High (85-95%) [66], [74], [76]	Low [74]	High [66]	Moderate [74]	Moderate to High [76]]
Social Network Analysis	Moderate to High (75-90%)[80], [82]	Moderate [82]	High [80]	Moderate [82]	High [80]

significantly enhanced the quality of the extracted features. These advanced methods capture semantic relationships, contextual nuances, and even cross-lingual patterns, greatly improving the accuracy and generalisability of fake news detection systems [81], [100]. Moreover, the integration of multimodal features, incorporating not only text but also user behaviour, social network structures, and even visual elements, has further enhanced detection capabilities [41], [43], [44], [76]. Ongoing research in feature extraction focuses on developing more robust, adaptable, and efficient techniques that can handle the dynamic nature of fake news, operate in real-time scenarios, and generalise across different domains and languages. Recent studies have explored complex features such as Zipf’s exponent and Multifractal Detrended Fluctuation Analysis (MFDFA) indices [15], as well as advanced semantic representations [34]. As misinformation tactics continue to evolve, the importance of sophisticated feature extraction methods to stay ahead of fake news propagation cannot be overstated. Emerging approaches, such as few-shot and zero-shot learning [47], [100] and explainable AI techniques [25], [44] are paving the way for more

adaptable and transparent fake news detection systems. Figure 8, presents a comprehensive overview of the various feature categories employed in the text analysis. It categorises the features into six distinct groups: Content-based, Semantic, Linguistic and Stylometric, User and Social Context, Complexity-based, and Multimodal. This structure offers a clear and organised visualisation of the feature landscape in the text analysis. This overarching meta-category serves as an integrative framework that unifies all subsidiary feature categories, facilitating a comprehensive understanding and multi-level analysis of features across the entire spectrum of detection methods.

1) CONTENT-BASED FEATURES

This category primarily focuses on features derived from the text itself, such as

- **Headline-body similarity:** This feature measures the similarity between the headline and the main body of the text, providing insights into content consistency and coherence [54], [80].

- **Source credibility:** This feature assesses the trustworthiness and reliability of the source of the text, which can be crucial for evaluating the validity of the information [72], [88].
- **Image features:** This category is considered in multi-modal approaches and is often extracted using deep learning models such as VGG [76]. It encompasses features extracted from images accompanying the text, such as visual content, colour schemes, or image composition, which can contribute to a holistic understanding of the message [2], [76].
- **5W1H components:** These features analyze the structural components of the text, such as (Who, What, When, Where, Why, How), which present the main elements of an article: fact, actor, place, time, reason, and manner [42].

2) SEMANTIC FEATURES

This category focuses on the meaning and interpretation of the text, including:

- **Time series:** A sequence of data points indexed in time order, typically collected at successive, equally spaced time intervals, used for analysing trends and patterns [7], [34], [43].
- **Named Entity Recognition:** This feature identifies and classifies entities such as persons, organisations, locations, and dates within the text, providing structural information about the content [21].
- **Sentiment analysis:** This feature analyzes the emotional tone and subjective opinion expressed in the text, revealing the writer's attitude and stance [12], [21], [29].
- **Topic modelling:** This feature identifies recurring themes and subjects within the text, offering a high-level understanding of the content [29], [88].
- **Word embeddings:** This category utilises techniques to represent words as vectors in a multi-dimensional space, capturing their semantic relationships and allowing for the computational analysis of meaning [17], [25].

3) LINGUISTIC & STYLOMETRIC FEATURES

This category focuses on the grammatical and structural aspects of the text, encompassing

- **Stylometric features:** This category analyzes the writing style of the text, including features such as sentence length, vocabulary richness, and use of punctuation, providing insights into authorship and writing style [23], [87].
- **Syntactic patterns:** This feature examines the grammatical structure of sentences, including dependencies, verb phrases, and noun phrases, revealing the underlying grammatical organisation [33], [42].
- **Readability indices:** This feature assesses the difficulty and complexity of the text, providing insights into its readability for different audiences [23], [29].
- **POS (Part of speeches) tags:** This category analyzes the parts of speech (nouns, verbs, adjectives, etc.) used in

the text, offering insights into the grammatical structure and function of words [10], [28].

- **LIWC (Linguistic Inquiry and Word Count):** A computerized text analysis method that categorizes words into psychological categories with special dimensions such as emotion, action, person, etc., producing scores used as features in text analysis tasks [98].
- **Character n-grams:** This feature analyzes sequences of characters within the text, providing information about word structure and potential spelling errors [19], [70].
- **Word n-grams:** This feature analyzes sequences of words within the text, capturing patterns of word usage, and revealing contextual relationships between words [19], [70].

4) USER AND SOCIAL CONTEXT

This category focuses on features related to the user who generated the text and the broader social context in which it exists, often analysed using graph-based methods and network analysis techniques, including:

- **User behaviour patterns:** This feature examines patterns in the user's behaviour, such as frequency of posting, engagement levels, and interaction with others, providing insights into their online activity [7], [69].
- **Audience demographics:** Statistical characteristics of a population or group, including age, gender, income, education, and location, are used to understand and categorise the target audience [43].
- **Social network information:** This category explores features derived from the social context of the text, such as the user's network connections, the platform on which the text was shared, and interactions surrounding the text [5], [43].
- **User profile feature:** This feature examines the characteristics and attributes associated with the user who generated the text, providing context regarding their background, interests, and writing style [7], [42], [69].

5) COMPLEXITY-BASED FEATURES

These features have been introduced to capture intrinsic textual complexity, which can differ between human-generated and machine-generated content [15], such as:

- **Zipf's Exponent:** This feature measures word frequency distribution in text, with deviations from expected patterns potentially indicating artificially generated or manipulated content, useful for distinguishing natural from synthetic text in fake news detection.
- **Multifractal Detrended Fluctuation Analysis (MFDFA) Indices:** This feature analyzes the multiscale structure and long-range correlations in the text, revealing subtle patterns that may differentiate genuine from fake news by quantifying text complexity and structure.
- **Recurrence Quantification Analysis (RQA) Indices:** This feature identifies repetitive structures and hidden patterns in text, potentially revealing the characteristics



FIGURE 8. Features Categories for Fake News Detection (FND): This diagram presents a comprehensive taxonomy of feature categories used in fake news detection models. The features are grouped into six broad categories: Content-based features, Semantic features, Linguistic and Stylometric features, Complexity-based features, Multimodal features, and User and Social Context.

of artificially generated content by measuring determinism, entropy, and laminarity in textual structure.

6) MULTIMODAL FEATURES

This category goes beyond text-based features and considers the integration of other modalities since it combines textual and visual information, such as:

- **Text-image coherence:** This feature analyzes the coherence and relationship between the text and accompanying images, assessing the effectiveness of visual elements in conveying meaning [2], [76].
- **Cross-modal representations:** This feature incorporates stylometric and semantic features, along with visual characteristics from images. These multimodal approaches aim to capture inconsistencies between text and associated images, which is a common characteristic of fake news [2], [39], [41], [76].

Table 7 provides an overview of the approaches, features, techniques, evaluation metrics, and limitations associated with these detection method.

The field of fake news detection techniques has evolved significantly, showcasing a diverse array of approaches ranging from traditional machine learning to advanced deep learning. A clear trend towards leveraging sophisticated language understanding capabilities is evident, with models such as BERT, CNN, LSTM, and Bi-LSTM frequently appearing in recent studies [8], [25], [80], [86], [102]. This shift is complemented by the rising popularity of ensemble methods

and hybrid models [26], [40], [41], suggesting that the combination of multiple techniques yields superior results. The classification features have also progressed, moving beyond simple lexical analysis to incorporate complex semantic, stylometric, and even multimodal representations that include images and user behaviour data [39], [41], [76]. This evolution reflects the field's response to the increasingly nuanced nature of misinformation.

The evaluation metrics used in the studies consistently included accuracy, precision, recall, and F1-score, facilitating standardised comparisons. Many reported results are impressively high, with several studies achieving accuracies of over 90% [8], [26], [80], [101]. However, it is crucial to note that these results are often dataset-specific and may not generalise well to real-world scenarios. The datasets employed vary widely, from general news corpora to specialised collections focusing on specific events, such as COVID-19 [27] or political news [16]. While this diversity is beneficial for exploring different aspects of fake news, it complicates direct comparisons between studies. Certain datasets, such as ISOT [45], [79], [80], [95] [45], [79], [80], and [95], FakeNewsNet [26], [44], and various Twitter-based collections [16], [28], [83], appear frequently, suggesting their emergence as standard benchmarks.

Despite these promising advancements, persistent challenges remain in the field of fake news detection. Common issues include the inability to handle context effectively [37], difficulties with multilingual or cross-domain

TABLE 7. A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

Ref.	Features	Techniques Used	Method Category	Dataset Used	Evaluation Metrics				Limitations
					Acc	P	R	F1	
[80]	Vector distances between titles and summarized contents using Cosine similarity, Jaccard distance, and Euclidean distance.	DNN, LSTM, GRU, BERT, ELECTRA-small (ProposedFE+DNN) based on Vector distances	DL	ISOT dataset, Fake-or-Real-News dataset, D3 (ISOT+ FRN)	100	98.55	99.94	98.79	Limited to datasets containing both 'content' and 'title'; may not perform well on skewed or different language datasets.
[41]	Stylometric and semantic features, visual characteristics from images.	SSM model , Hyperbolic Hierarchical Attention Network (Hype-HAN), Error Level Analysis (ELA), pre-trained EfficientNetB7	DL Comp. Vision	FNDD, TI-CNN, FNSDS	96.90	98.88	95.68	97.25	Domain-level error analysis, ground truth source-level labels, weak supervision, limited resources.
[25]	Word embeddings using a BERT-based model.	BERT, SHAP (SHapley Additive exPlanations)	DL, AI	MMCovid19 dataset	92.6	92.7	92.6	92.4	Limited to simple adversarial attacks, heavily disturbs original content.
[7]	Global vectors (GloVe), deep learning features.	CNN, DNN, RNN (RNN+GloVe)	DL	Open-source dataset of millions of news articles	98.99			98.97	Limited to English tweets, inability to understand language with different meanings.
[85]	Word2vec embeddings, TF-IDF, GloVe embeddings.	SSPN (Simple Shallow Perceptron Network) ANN, CNN, RNN, SVM, Logistic Regression, BiLSTM, RNN-GRU, LSTMs	ML, DL	LIAR dataset, FakeNewsNet dataset	60.00	58.0	97.0	73.0	Inability to distinguish between fake and real news/data, curse of high dimensionality, higher training time, over-fitting.
[68]	Creator-based, content-based, context-based features.	NLP techniques, supervised and unsupervised classification	ML, NLP	Twitter datasets	80.0			78.97	Not explicitly mentioned.
[7]	Metadata, content, interaction, community, account-based, and content-based features Linguistic Regression.	KNN, Decision Tree, Gaussian Naive Bayes, SVM, Random Forest , AdaBoost, XGBoost, RNN, LSTM, BiLSTM, CNN	ML, DL	Dataset by Cresci et al., Social HoneyPot Dataset	Accuracy: LR-RF: 99.06, AB& SVM: 99.01. All features: AB:98.98 Precision: K-NN: 98.16, ANOVA F-value (SVM) & LR (SVM, RF, K-NN): 98.0. Recall: AB, ExtraTrees & LR: 99.73				Imbalanced dataset, the computational expense of feature extraction.
[69]	Number of friends, followers, sentiment score, h-index score, influence score.	Random forest , multilayer perceptron, support vector machine	ML	Dataset of 50,000 politicians on Twitter	96.0	94.0	97.0	95.0	Not explicitly mentioned.
[86]	Lexical and sentiment features, n-gram features, Empath generated features.	Naïve Bayes, SVM, LR, CNN, K-NN, LSTM, Bi-LSTM, C-LSTM, HAV, BERT , RoBERT, DistilBERT, ELECTRA, ELMO	ML, DL	LIAR dataset, Fake or Real dataset, Combined Corpus dataset	96.0				Limited improvement on certain datasets, inability to handle informal language.
[87]	Convolutional neural network (CNN) model features.	CNN tailored for small text messages	DL	Twitter dataset	95.03				Not explicitly mentioned.
[12]	Bot scores and sentiment values.	Decision Tree Algorithm, TweetNLP approach, RoBERTa, XLM-R	ML NLP	Dataset from Twitter	Not provided				Purely observational relationships, lack of causation.
[14]	Linguistic semantic features, data mining, knowledge graph, multimodal representation learning.	Boosting with Transformers Encoder, tree structure, sequence network, graph neural network, multimodal graph convolutional networks	ML, DL	Out-of-Context Detection Dataset in COSMOS	89.75	86.22	94.68	90.53	Difficulty in distinguishing between out-of-context and not-out-of-context captioned images.
[15]	Zipf's exponent, Multifractal Detrended Fluctuation Analysis (MFDFA) indices, Recurrence Quantification Analysis (RQA) indices.	RBF-Support Vector Machine (SVM), Genetic algorithm	ML	Corpus of human-generated and machine-generated texts	93.58				Complexity indices may not be sufficient to discriminate between human- and machine-generated texts.
[16]	Tweet text, number of likes, retweet count, user ID, user name, user	Sentiment Lexicon (Multinomial Naive Bayes classifier)	ML, NLP	Twitter dataset related to the 2020 U.S. presidential election	89.9				Analysis based only on English tweets, geolocation limited to state-level.

TABLE 7. (Continued.) A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

[17]	join date, user followers count, user location. User profile features, Word2Vec features, BERT features.	Transformer with Graph convolutional networks (GCNs-TM), SAGE-MT and GAT-MT recurrent neural networks (RNN), convolutional neural networks (CNN), generative adversarial networks (GAN)	DL	CoAID dataset, Politifact dataset, Gossipcop dataset, weibo	99.77	98.78	98.90	98.66	Not explicitly mentioned.
[70]	Word n-grams, character n-grams, pre-trained embeddings.	Linear SVM, the neural network-based model with attention mechanism, mBERT, XLM-R	ML, DL	SemEval 2016 stance dataset, multilingual TW-10 corpus IberEval 2018, CIC-CA & CIC-Random		mBERT RT XLM-R	CIC Random CIC Random	74.08 72.79 74.68 74.87	Small size and skewed class distribution of resources.
[88]	Post-level features, user-level features, sentiment analysis features.	CerCDA , CNN-LSTM , Deep learning framework, sentiment analysis model.	DL	PHEME dataset	81.0	79.0	79.0	79.0	Lack of integration of hashtags as microblog features, failure to consider semantic relationships.
[8]	Word embeddings, linguistic features.	BERT, BiLSTM, Covid-twitter-BERT (CT-BERT), RNN and BiGRU, BiGRU on top of the CT-BERT	DL	COVID-19 related social media dataset	98.5	97.97	99.11	98.5	Limited exploration of hyperparameters and optimization techniques, relatively small COVID-19 dataset.
[9]	Bag-of-words approach.	SMOTE and classifier vote ensemble (SCLAVOEM) method, hyper parameter optimization approaches (NB, SMO, VP, KNN & Random Forest (RF)	ML	COVID-19-related Twitter datasets	99.66	99.9		99.7	Concentration of tweets from a single country, English language limitation.
[18]	Word embeddings using a BERT-based model.	Adversarial Training (AT) by using T5 generative it, PLMs such as BERT	DL	Xsum, SQuAD, Writing, IMDP, HC3	96.01	-	AVG -	95.93	Limited to simple adversarial attacks.
[89]	News articles feature.	Easy Data Augmentation (EDA) techniques, CNN (CNN-EDA)	DL	Jigsaw toxic comments dataset, fake news dataset	92.0 98.0	74.0 96.0	0.85 89.0	78.0 92.0	Limited improvement in performance on certain datasets.
[72]	Credibility scores of publishers, users, messages, and headlines.	Integrated deep CNN Bi-LSTM (N-DCBL) with TF-IDF_Glove , attention network, deep learning-based multiheaded attention network, ResNet	DL	Buzzface, FakeNewsNet, Twitter	99.93 99.67 99.97	96.96 97.96 98.96	95.69 96.69 97.69	98.78 99.08 98.08	Not explicitly mentioned.
[19]	Grammatical, orthographic, text measure, and linguistic features from Headline (Post Text), Title (Target Title), and Body Content (Target Paragraph) with Concatenated and Average embedding vector.	SVM with concatenated S-BERT embedding (DBCD)	ML, DL	Webis Clickbait Corpus 2017, dataset by Chakraborty et al.	84.0 98.0			82.0 98.0	Only considers the English language dataset.
[20]	Linguistic features based on the LIWC (Linguistic Inquiry Word Count) program.	Extreme gradient boosting (XGBoost) algorithm	ML	Dataset of 1124 health-related news stories and press releases	-	89.78	76.88	82.83	Limited content of analyzed articles.
[21]	Sentiment features, linguistic features (LFS), named entity-based features.	Gated Recurrent Units (GRU), Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN)	DL	COVID-19 Fake News dataset	86.12	-	-	-	-
[90]	Text-specific features, user-specific features, message-specific features (Writing pattern, Grammar, Psycho-Linguistics and Quantity).	Naive Bayes, Support Vector Machine (SVM), Decision Tree, CNN, Bert, Random Forest (RF), K_Nearest, Neighbor, Boosting, WELFake using word Embedding.	ML, DL	WELFake dataset	96.73	-	-	-	Focused on analyzing writing patterns. Not extended to other factors like knowledge graphs and user credibility.
[23]	Complexity features, readability index, psycholinguistic features, stylistometric features.	Gaussian Naïve Bayes (GNB), Logistic Regression (LR), Random Forest (RF), Linear Support Vector Classifier (SVC), K-nearest neighbour (KNN)	ML	Mc_Intire dataset Reuter dataset Buzzfeed Random Political News,	90.8 90.0 93.8 86.9	- - - -	- - - -	- - - -	Not explicitly mentioned.

TABLE 7. (Continued.) A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

[24]	Doc2Vec Bag-of-Words features.	PU-LP algorithm	ML	Datasets in Portuguese and English, FakeNewsNet, Fake.BR, Fake-checked News	92.76	Need for thresholds to be set for classifying examples.			
[26]	Writing style features, news content features, user-related features, and publisher-related features. Credibility, Influence, Sociality, Validity and Lifetime.	Novel sentence-level convolutional neural network (SLCNN), CreditRank algorithm Multi-model Framework (FR-Detect)	DL	A Benchmark dataset called FakeNewsNet was collected from (GossipCop+ PolitiFact)	99.0	98.0	98.0	98.0	Incomplete multi-modal datasets.
[27]	Linguistic features, social contextual features.	Logistic model tree, stochastic gradient descent, classification via clustering, bagging, decision tree, RNNs, CNNs, BERT, DistiBERT & SHAP	ML, DL	Covid-19 AAI2021-Covid-19	97.2	93.75	93.85	93.8	Limited dataset collected by April 2020.
[73]	ID of content, full textual content, 126 additional features.	Dark Crawler, Posit Toolkit TensorFlow (Deep Neural Networks), Random Forest , LibShortText, LibLinear.	ML, DL	Dataset of 6,562,080 'real news' and 'fake news' items	81.60	Twitter+ posit & Charcount feature	Limited content of Russian IRA tweets.		
[92]	Novelty and emotion representations of contents and comments.	Multilayer Perceptron (MLP), BiLSTM model (Multitask Framework) PMTM: Proposed Multitask Model	DL	Pre-trained: Quora Question Pair Dataset, Unified Dataset, GoEmotion Dataset, The target for the main task: ByteDance, FNC-1, Covid-Stance, FNID datasets	78.0	Twitter+ posit feature only			
					76.50	Facebook+ posit features only			
					74.40	Twitter+ Charcount only			
					73.50	Facebook+Posit+Charcount			
[93]	Linguistic characteristics, n-grams Data Privacy along with Similar accuracy, Lesser Losses and Low Processing Resources.	Federated-Fake News Classification (F-FNC) LSTM, Bi LSTM, CNN-LSTM, CNN-Bi LSTM	ML, DL	Independent & Identically Distributed (IID) and Non-IID datasets	98.19	ByteDance dataset	-		
					96.87	Fake News Challenge-1 (FNC-1) dataset			
					96.07	Covid-Stance dataset			
[74]	Textual and visual components, Word2Vec: Various word vector representations.	Pre-Tr-ConvRNet-De-Fuz-Lear classifier, De-Fuz_lear, DNN (Deep Neural Network), CNN (Convolutional Neural Network), LR (Logistic Regression)	DL	Kaggle (Social Media dataset)	98.86	Fake News Inference Dataset (FNID)	Need for mapping available federated learning programming libraries.		
					90-92	f-FNC Multi-CLient			
					91-92	f-FNC Sigle-CLient			
[29]	Content Feature, Textual, sentiment, semantic, and readability features.	Random Forest, Support Vector Machine (SVM), Decision Tree, LightGBM, XGBoost, AdaBoost. Stacking-Ensemble Learning	ML	The dataset contains 24,517 news from fact-checking. All Data (Politifact+Snope+Truthorfiction) Multi-Class Form, Binary Form	91-92	Traditional Distributed DL	Need for future improvements in automatic fact-checking tool.		
					90-92	Traditional Distributed DL			
[30]	Linguistic features, content-based features, word embeddings.	Logistic Regression, RNN, LSTM, GRU, RCNN with specific word2Vect Embedding trained , Attn+LST, CNN.	KL, DL	FakeNewsCorpus	95.46	96.73	94.87	95.46	-
[31]	Social media posts and user behavior.	Deep neural network architecture based on transformers	DL	CoAID dataset	83.60	83.97	81.94	82.81	-
[10]	Textual Content, BoW+ Metadata (number of characters, POS tags, publisher info).	Decision Tree (DT) , k-Nearest Neighbors (KNN), Logistic Regression (LR) , Linear Support Vector Machine (LSVM), Multinomial Naïve Bayes (MNB), Bernoulli Naïve Bayes (BNB), Perceptron, Neural Network (NN) , Ensemble Random Forest (ERF), Extreme Gradient Boosting classifiers (XGBoost)	ML	Ground-Truth dataset	83.97	91.52	93.21	93.65	93.43
					91.52	93.21	93.65	93.43	Limited exploration of data augmentation techniques.
					91.52	93.21	93.65	93.43	
					91.52	93.21	93.65	93.43	
					87.56	87.55	88,11		Sample size less than 3,000 users, final f-score below 0.6
					87.56	87.55	88,11		
					87.56	87.55	88,11		
					F1 score: 0.8076				Ground-truth data is expected to change continuously.
					F1 score: 0.8076				
					F1 score: 0.8076				
					93.75	KNN			
					93.75	KNN			
					93.75	KNN			
					99.68				
					99.68				
					99.68				
						99.74	99.93	DT	
						99.74	99.93	DT	
						99.74	99.93	DT	
								LR	
								LR	
								LR	

TABLE 7. (Continued.) A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

[32]	Linguistic features, content structure, global semantic relationship, local sequential order.	<u>SemSeq4FD Model, SA-GN</u> , Graph convolutional neural networks (GCN), LSTM, SVM, logistic regression, CNN,	ML, DL	LUN, SLN, Weibo, RCED datasets	93.78 88.42	93.90 89.04	93.78 88.42	93.78 88.38	Need for more detailed research to improve the representation effect.
[33]	Linguistic properties of text (language complexity, syntactic patterns, word categories).	Linear method trained on stylistometric features, recurrent neural network with tuned BERT	ML, DL	NewsStyleCorpus 2	Doc.c v 86.80 90.50 96.29	Sur. cv 8081 87.32 92.92	Accuracy Stylometric Neural BERT		Human annotators had access to full contexts when doing their task.
[34]	TF-IDF representations, time-series data, word2vec, node2vec, BERT-based representations	Generic Deep Markov Random Field Nural Network (GDMFN), Word2vec, doc2vec, transformer-based models, graph convolutional neural networks, mean-field layers	DL	Twitter Weibo PHME	79.3 96.4 -	79.8 96.4 67.4	79.2 96.4 67.6	79.3 96.4 66.7	Performance may vary at different runs due to random initialization.
[75]	Embedded representations of web page URL, content, and DOM structure, URLs in the page.	CNN-BiLSTM network with an attention mechanism VecCred (a-c-d-e)	DL	Microsoft credibility dataset(D1), Medical web reliability corpus(D2), CLEF eHealth 2020 task-2 (D3) D3(BI) Binomial Intervals	90.47 99.71 82.56 82.0	94.21 99.71 83.11 81.0			Only takes into account information directly extractable from the web page.
[95]	Content-free, content-specific, stylistometric, and folksonomy features.	<u>Transformer-based, RNN, CNN</u> , BERT, LSTM.	DL, Ensemble Methods	GRAFN ISOT		88 Upto 99	77 Upto 99	82 up to 99	Performance deterioration when exposed to news samples from different domains. Biases in the dataset used generalizability resulting in the need for further validation of diverse datasets.
[96]	CNN model features.	Deep convolutional neural network (FNDNet)	Deep Learning	Kaggle fake news dataset	Accuracy: 98.36%				It needs to extend to multitasking fake news detection and including multimedia for analysis.
[81]	Novelty and emotion-aware representations of news headline-body pairs.	FND Model, Logistic regression, deep neural architectures, transfer learning-based models	ML, DL, Transfer Learning	Quora Question Pair Dataset, Unified Dataset, ByteDance Dataset , FNC-1, Covid-Stance, LIAR-PLUS	ACC=78.3-96.11				Imbalanced datasets & potential classification bias.
[76]	Textual information and images.	CNN, BERT , BiLSTM A Fine-grained classification	DL	Multi- Fakeddit dataset Uni- Fakeddit dataset	87 78				
[97]	Headlines, body text, images/videos, news sources, writing style.	Ensemble Learning model, LSTM, Bi-LSTM, CNN , the hybrid model of (CNN and LSTM)	DL	COVID-19 Fake News dataset	94.24	93.6	93.9	93.7	Not explicitly mentioned.
[98]	Word embedding, grammar analysis, LIWC text analysis, uni-grams and bi-grams.	Ensemble Learning model , Embedding LSTM, depth LSTM, LIWC CNN, N-gram CNN	DL	Fake_or_Real_news, Snopes_Fake_Legit_news, Fake News Detection datasets	72.3	The cross-domain intractability issue up			Only considers main texts in news, not suitable for early detection.
[35]	Numerical and categorical features (42 in total).	Support Vector Machines (SVM), Random Forest, Logistic Regression, <u>Decision Tree</u> , Multi-layer Perceptron (MLP), Adaptive Boosting (AdaBoost), Gaussian Naive Bayes (GaussianNB), pre-trained transformer models, BETO model with RUN-AS feature	ML, DL	Spanish dataset focused on health and COVID-19	95 88.7	-	-	94.8 85.4	Complexity of annotating semantic elements.
[36]	Features based on the MbMix method.	BERT pre-trained language model, MbMix	DL	RTE, MRPC, SST-2, IMDB, TREC-Fine , QNLI, OLID, COLA datasets	93.6				-
[37]	Syntactic and semantic information from text, word embedding (Word2Vec & BERT).	Dense, CNN, LSTM , BiLSTM models, LSTM with mimicked in Threat, hate & Insalt.	DL	Kaggle challenge on toxic comments dataset	AVG	91.86	95.2	93.4	Models trained without context, and figurative language might mislead models.
[99]	Toxicity-based features (toxicity, severe	Linear SVM, Random Forest, BERT-based variants	ML, DL	GitHub dataset of false and real	92	98.3	88.4	95	Basic deep learning techniques not

TABLE 7. (Continued.) A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

	toxicity, obscene, identity attack, insult, threat, sexually explicit).	(BERT_SVM, BERT_RF), LSTM NN-based models		tweets, Facebook posts, and news articles about COVID-19					explored, the potential for enhanced performance.
[100]	Features extracted from news articles and news bodies.	Multilingual Bidirectional Encoder Representations from Transformer (BERT) model	DL	A dataset of 4,900 news items from various domains and languages	83.54				Poor performance in specific domains, vulnerability to certain types of news headlines and bodies.
[39]	Textual information and images.	Ensemble deep learning model, Visual Bidirectional Encoder Representations from Transformers (V-BERT)	DL, Ensemble Methods	Fakeddit dataset	88.83	93.54	90.29	91.89	Potentially incorrect labelling of news, choice of assessment metrics.
[40]	TF-IDF, Bag of Words (BoW), Word2Vec.	Hard voting ensemble (PHVE), oft voting ensemble model (PSVE), (GloVe + Conv1D + Bi-GRU + Bi-LSTM) with Self-attention mechanism (GCB-SA) and (FastText + Conv1D + Bi-GRU+ Bi-LSTM) with Self-attention mechanism (FCB-SA). Multinomial Naïve Bayes (MNB), Logistic Regression (LR), Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting Trees (GBT), CNN, LSTM, Self-Attention Mechanism.	ML, DL	SMS spam Ling spam, Twitter Fake news datasets	91.65 88.97	91.67 88.98	91.65 88.97	91.63 88.92	Lack of large-scale publicly available benchmark datasets.
					93.09	93.10	93.09	92.99	
					90.05	90.08	90.05	90.04	
					93.85	93.82	93.86	93.87	
					95.76	95.76	95.76	95.76	
[82]	88 features organized in surface information, part of speech (POS), discursive characteristics, readability indices.	XGBoost, SVM (radial and linear), decision tree, Naïve Bayes	ML	Job News database with 42,677 news articles: ISOT, COVID-19 Fake News Dataset	99.9 96.33				Difficulty in selecting and classifying appropriate linguistic features.
[83]	Chi-square test, keywords, Word Sketch.	RoBERTa, DistilBERT, BERT, XLNet, ELECTRA, Longformer, CNN , MultiCNN, Bayes, SVM	ML, DL	PFNSoT dataset: (Spanish and English tweets)	71.5 81.5				The dataset is limited to the last 100 tweets of each user.
[42]	Content-based features, context-based features, 5W1H components.	FNDeepML: Logistic regression models , DT, SVM, Naïve Bayes (multinomial), 5W1H.	ML, DL	Spanish dataset (491 true news and 480 fake news)	80.7			97.8	Classification limited to True/False/Unknown categories
[43]	News content, social media content, user profiles, audience demographics.	Graph convolutional networks (GCNs), RNNs, CNNs, Graph Global Attention Network with Memory (GANM): GAT-GANM, GCN-GANM , SAGE-GANM	DL	Politifact dataset Weibo dataset Gossipcop	98.35 98.25			98.35 98.25	Struggles with multi-source, multimodal, data and complexities in distinguishing fake news from genuine news
[44]	Linguistic indicators, social context information, user interactions, news content.	Recurrent neural networks (RNNs), co-attention-based techniques	DL	FakeNewsNet dataset: (Politifact + Gossipcop)	93.6 84.3	93.8 80.5	97.7 81.9	95.1 79.7	Absence of fact-checking websites in learning mechanism
[77]	Text-based features (hashtag count, noun count, headline length, etc.), social cyberspace features, user profiling, linguistic and syntactic.	AWD-LSTM model, various machine learning algorithms (DT, RF, KNN, Bayes Network, Naïve Bayes, SVM)	ML, DL	45 datasets (28 Canada-related, 22 with hostile activities): (Real Twitter (DS18) & Spmobot2)	98.93	98.80	99.10	98.90	Lack of labelled dataset for fake/real news articles, data access restrictions and dynamic nature of streaming data.
[101]	Token sequences from the dataset are transformed into vector representations using BERT Tokenizer.	Pre-trained BERT models	DL, ML	Kaggle dataset (fake and true news files)	99.96	99.96	99.96	99.97	Use of small dataset, lack of consideration for the context of statements.
[45]	Features based on analysis of large material of fake and real news.	BERT , RoBERTa , CNN . Hybrid architectures (-DRE & TDE)	DL, ML	ISOT dataset, COVID-19 dataset	F1-scores above 90				Truncation of input strings to 100 tokens due to BERT limitations.
[46]	Lexical and semantic features.	Multi-layer perceptron (MLP), convolutional neural network (CNN), recurrent neural network (RNN), hybrid CNN-RNN	DL	ISOT dataset, LIAR dataset, Kaggle fake-news dataset	--				Lack of cross-dataset analysis, and need for data augmentation techniques.
[47]	Features generated by representing news articles as finite metric	Recurrent neural networks (RNNs), BILSTM , BERT , ensembles with TDA	DL, Topological Data Analysis	The dataset contains 21,417 real news and	Acc= 92-99				A study based on a specific task of fake news identification, may not apply to

TABLE 7. (Continued.) A summary table of the approaches, features, techniques, evaluation metrics, and limitations in the selected studies.

[103]	spaces using GloVe embeddings. Content (Text) of News articles.	1-S FND: Semi-Supervised Learning, Adaptive Pseudo-Labeling for unlabeled data, Gradients-based interpretation for feature extraction, utilization of transformers and BERT for FND.	DL	23,481 fake news articles (Kaggle) FND1 (2017) , GossipCop (2019), Politifact (2019), FND2 (2018)	98.75	98.75	98.75	98.75	other text classification tasks. Potential generalization issues and unspecified computational resource requirements.
					98.57	98.58	98.57	98.57	
[102]	TF-IDF vectorization, URL validation, grammar checking, swear word detection.	BERT , Machine learning classifiers, Manual program-based approach	DL, ML	Datasets from Politifact, Channel4, Snopes, social media, and Kaggle (English tweet)	Accuracy > 90				Limited discussion on ethical considerations and comparative analysis.
[78]	Contextual word embeddings, metadata (subject, speaker's info, political affiliations).	BERT (triple BERT network) , CNN, Bi-LSTM, Ensemble networks	DL, ML	LIAR Binary label, LIAR Six-PLUS label	74.0	69.9	85.4	76.8	Model complexity, potential lack of interpretability.
[79]	Summarized feature vector representation.	Bi-LSTM , CNN-based attribute-wise attention, GloVe word embeddings	ML, DL	Kaggle_fake_real_ne-ws (2022 & 2016), ISOT fake news dataset, LIAR dataset	92.92	92.97	92.90	92.90	Need for multi-modal and cross-lingual approaches, potential dataset bias.
					99.82	99.82	99.81	99.81	
					99.87	98.63	98.63	98.63	
					60.31	60.61	60.63	60.31	

generalisation [95], [100], and struggles with subtle forms of misinformation. Many studies acknowledge the need for larger, more diverse datasets [77] and the challenges of keeping pace with evolving disinformation tactics. Concerns about potential classification bias [76] and vulnerability to certain types of news headlines and bodies [100] are recurrent themes. Furthermore, the increasing complexity of deep learning approaches has sparked calls for more interpretable models [42], as understanding the rationale behind the classifications has become crucial. These limitations highlight important directions for future research, including the development of more robust, generalisable, and interpretable fake news detection systems, the need for automatic fact-checking tools [74], and improved handling of multi-source multi-modal data [43].

D. DATASET

In this subsection, we present an overview of the datasets used in the reviewed studies, as summarised in Table 8. These benchmark datasets have been widely adopted for training and evaluation purposes in fake news detection research. Our systematic review revealed the evolving landscape of datasets employed in machine- and deep-learning approaches for fake news detection. These datasets have progressively diversified in size, modality (i.e. the type of data they contain, such as text, images, or user interaction data), and focus, reflecting the multifaceted challenges inherent in combating online misinformation [47], [80], [86].

Many studies are based on large-scale, publicly available datasets such as LIAR, Fakeddit, FakeNewsNet, and ISOT [26], [28], [39], [76], [82]. These datasets typically contain thousands to millions of labelled news articles or social media posts, providing a robust base for model training and evaluation. Notably, some datasets, such as FakeNewsNet and PHEME, go beyond mere text content to include

social context information, such as user engagement data and propagation networks [26], [88]. This additional layer of information enables researchers to explore the integration of user behaviour and social network dynamics with textual features. For instance, studies utilising these enriched datasets have demonstrated improvements in fake news detection accuracy of up to 14.5% compared to models using only textual data [25], [72]. Building on these foundational datasets, researchers have developed specialised datasets to address emerging challenges. Domain-specific datasets, particularly those focused on COVID-19 misinformation, such as CoAID and MMCovid19, have become increasingly important in tackling timely and potentially harmful misinformation [8], [21], [25], [97]. These datasets often feature multilingual content, facilitating the investigation of transfer learning techniques and methods that apply knowledge gained from one language to improve detection in others and enhance fake news detection across different languages [94]. Other specialised datasets cover areas such as health misinformation [20], political news [16], and celebrity gossip [26], each addressing unique challenges in their respective domains. To tackle the challenge of detecting novel forms of fake news, some researchers have created datasets suitable for few-shot and zero-shot learning approaches. These methods allow models to identify fake news with minimal or no prior examples, as demonstrated by the Fakeddit dataset with its diverse range of fake news categories [39], [76], [100].

Furthermore, the push for more transparent and explainable AI in fake news detection has led to the development of datasets that include not only labels but also explanations or justifications for the classifications [56], [81]. This trend addresses the “black box” problem, which is often associated with complex AI models. Examples include the LIAR-PLUS dataset, which extends the original LIAR dataset with fact-check justifications, and the PubHealth dataset, which

TABLE 8. Summary of publicly available fake news datasets with key attributes and source URLs.

Dataset	Modality	Size	Labels	Type	URL & Ref
LIAR	Text	12,836 statements	6 truthfulness levels	Political statements	politifact.com [13], [46], [77], [78], [85], [86], [102]
FakeNewsNet	Text, Social context	Varies	Real, Fake	News articles	github.com/KaiDMML/FakeNewsNet [26]
Fake-or-Real-News	Text, Social context	6335 articles	Real, Fake	News articles	https://www.kaggle.com/datasets/jillanisoftech/fake-or-real-news [80]
ISOT	Text	44,898 articles	Real, Fake	News articles	uvic.ca/engineering/ece/isot [79], [80], [82]
CoAID	Text, Social context	~4,000 fake, ~7,000 real	Real, Fake	COVID-19 news	github.com/cuilimeng/CoAID [13], [31]
Gossipcop	Text, Social context	22,140	Fake, true	News articles	https://github.com/KaiDMML/FakeNewsNet/tree/master/dataset [13], [26], [44]
MMCovid19	Text (Multilingual)	~11,000 documents	Real, Fake	COVID-19 news	github.com/bigheiniu/MMCoVaR [25], [94]
Fakeddit	Text, Image	Over 1 million	2-way, 3-way, 6-way	Multi-modal fake news	github.com/entitize/fakeddit [39], [76]
LIAR-PLUS	Text	12,836+ claims	6 truthfulness levels	Political statements with justifications	github.com/Tariq60/LIAR-PLUS [78], [81].
WELFake	Text	72,134 articles	Real, Fake	Merged news dataset	https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification?resource=download [90].
PubHealth	Text	11,832 documents	True, False, Unproven, Mixtur	Public health news	https://huggingface.co/datasets/bigbio/pubhealth [94].
Multilingual Multidomain	Text	4,900 news items	Fake/Legit	News in multiple languages	http://www.iitp.ac.in/~ai-nlp-ml/resources/data/TALLIP-FakeNews-Dataset.zip [100]
Wikipedia	Comments	200,000	6 toxicity + non-toxic	Online textual	https://www.kaggle.com/ [37]
Health and COVID-19	Text	10,700 fake and genuine	Annotated structure	Health news	https://github.com/marionieto51/NewsReliabilityAnnotation [35], [97]
Health-related News	Text	1,124 stories		Health news and press releases	HealthNewsReview.org [20]
Weibo	Text	7300 articles	Real, Fake	News from 8 domains	https://www.datafountain.cn/competitions/422 [32]
LUN, SLN	Text	LUN=25542 SLN=360	Real, Fake	News, social media posts	https://homes.cs.washington.edu/~hrashkin/fact [32]
Twitter Dataset	Text	130 million tweets		Social media posts	https://victoriarubin.fims.uwo.ca/research/projects/data-to-go/
Webis Clickbait Corpus 2017	Text	22,033, 32000 samples	Clickbait, Non-clickbait	News headlines and content	https://github.com/theocjr/twitter-reader [87]
COVID-19 Misinformation	Text	10700 claims	Real, Fake	COVID-19 news	webis.de [19]
SemEval 2016	Text	4870 English tweets	AGAINST, FAVOR, NONE	Stance detection	https://github.com/parthpatwa/covid19-fake-news-detection [27]
FND1 (2017), FND2 (2018)	Text	4009, 20800	Real, Fake	Attributes	alt.qcri.org/semeval2016 [70]
Twitter Politicians	Text, User data	50,000 users	Trusted, Untrusted	User credibility	https://www.kaggle.com/jruvika/fake-news-detection
Cresci et al. Bot Dataset	Text, User data	3474 genuine, 4912 bot accounts	Genuine, Bot	Bot detection	https://www.kaggle.com/c/fake-news/data [103]
FNDD, TI-CNN, FNSDS	Text, Image	20,015, 45000	Real, Fake	Multimodal news	N/A [69]
Temporal Tweet Dataset (pHEME & pHEME-r)	Text, Temporal data	6424 tweets	Fake and non-fake	Temporal analysis	https://github.com/mpstewart1/AC209a-Twitter-Project/blob/master/Data_acquisition.md#1-cresci-2017-dataset [7]
IEO Twitter Dataset	Text	110,000+ tweets		Cryptocurrency news	https://drive.google.com/file/d/0B3e3qZpTccsMFo5bk9Ib3VCc2c/view [41]
GPT-2 Generated Corpus	Text	17028.3 tokens per text	Human, AI-generated	AI-generated text	https://figshare.com/articles/dataset/PHEME_dataset_for_Rumour_Detection_and_Veracity_Classification/6392078 and https://figshare.com/articles/dataset/PHEME_dataset_of_rumours_and_nonrumours/4010619 , [68], [88]
GoEmotion Dataset	Text	58,000 comments	27 emotion categories	Emotion-labeled data	N/A [12]
2020 U.S. Election Twitter Dataset	Text	1,747,805 tweets	Hashtags related to the US	Election-related tweets	https://www.gutenberg.org/ , https://github.com/openai/gpt-2 [15]
Microsoft Credibility Dataset	Text	1,000 web pages	5-point credibility scale	Web page credibility	https://github.com/google-research/google-research/tree/master/goemotions [92]
PFNSoT	Text (Spanish, English)	30,000+ tweets	Fake news, non-fake spreader	Multilingual fake news	https://www.kaggle.com/datasets/manchunhui/us-election-2020-tweets [16]
Spanish Fine-grained Dataset	Text	971 news items	True, Fake	Fine-grained annotated news	N/A [75]
OpenSources dataset	Text, tweets	Millions articles & 79000 propaganda	Fake, Real, Propaganda	1001 domains listed	https://github.com/marco-siino/fake_news_spreaders_detection [83]
Dataset (COSMOS)	Text, image	161,752 images, 41,006 captions	19 categories	Out-of-Context Detection	https://github.com/livinglang/fake_news_dataset/tree/1.0.2 [42]

TABLE 8. (Continued.) Summary of publicly available fake news datasets with key attributes and source URLs.

Dataset	Modality	Size	Labels	Type	URL & Ref
Canadian datasets	Text, posts	45 datasets (28 Canada, 22 hostile activities)	Fake, Real	Canadian news and hostile activities	https://www.cbc.ca/news/technology/quebec-fake-newspapers-1.5228905 [28]
Multi-source Multi-class Dataset	Text	23,935 news items	5 truthfulness levels	Recent multi-class news	Collected from: (PolitiFact.com, Snopes.com, and truthorfiction.com) [29]
Large-scale Multi-class Dataset	Text	100,000 articles	10 classes	Multi-class fake news	https://github.com/several27/FakeNewsCorpus [30]
Interactive Credibility Dataset	Text	103,219 document URLs were collected	Credibility ratings non-credible or credible	Interactive assessment	N/A [33]

provides explanation text fields alongside its labels [81], [94]. These enriched datasets enable researchers to develop models that not only detect fake news but also provide reasoning for their decisions, enhancing trust and interpretability in AI-driven fake news detection systems. Moreover, researchers have created datasets to address specific aspects of fake news detection, such as stance detection [70], user behaviour analysis [69], bot detection [7], and multimodal fake news identification [41]. These specialised datasets, along with efforts to create more comprehensive collections, such as the WELFake dataset [90], reflect the evolution of the field towards more sophisticated, context-aware, and adaptable approaches to combating online misinformation. However, it is important to note that challenges remain, such as keeping datasets current with rapidly evolving misinformation tactics and ensuring they represent diverse cultural and linguistic contexts.

In summary, the diversity and specialisation of datasets in fake news detection research has significantly contributed to the advancement of this field. By providing rich, varied, and increasingly nuanced data, these datasets enable researchers to develop more accurate, robust, and explicable models for detecting fake news. As the challenge of online misinformation continues to grow and change, the ongoing development of innovative datasets will remain crucial for our ability to effectively combat this global issue.

E. COMPREHENSIVE ANALYSIS OF RECENT RESEARCH GAPS IN FAKE NEWS DETECTION

A systematic literature review of fake news detection has revealed several critical research gaps that must be addressed to advance the field. These gaps span various aspects of fake news detection, including accuracy, multimodal analysis, security, real-time detection and cross-domain applicability. By examining these gaps and proposing suitable techniques and models, as shown in Table 9, we can chart a path forward for more effective fake news detection systems.

One of the primary challenges in fake news detection is achieving high accuracy, particularly in complex scenarios [23], [72], [80], [90]. Existing methods often struggle to capture the nuanced nature of fake news, which leads

to suboptimal performance. To address this issue, our study proposes hybrid models that combine linguistic features, semantic analysis, and advanced machine learning algorithms. BERT-based models with attention mechanisms show promise in this area, as they can capture complex linguistic features and contextual information, and their ability to understand the nuances of language and context makes them particularly well-suited for improving the accuracy of fake news detection. Building upon this foundation, another significant gap lies in multimodal analysis [41], [76], [96]. Many current approaches focus solely on textual content, overlooking the rich information available in images and videos that often accompany news articles. Implementing Stylometric and Semantic similarity-based multimodal fake news detection (SSM) can help capture the similarity of diverse modal data in multimodal news. Vision-language pre-training (VLP) models and Multimodal Transformers (e.g., VisualBERT, CLIP) are particularly suitable for this task, as they can effectively process both text and visual data simultaneously. By integrating these multimodal approaches, we can develop more comprehensive and accurate fake news detection systems that consider the full context of the news content. Although improving accuracy and expanding analysis capabilities are crucial, the security and robustness of fake news detection systems present critical challenges [25].

The potential security implications of using explainable AI (xAI) in fake news detection have been identified, with concerns that these systems could be vulnerable to adversarial attacks. To enhance the robustness of xAI systems, this study suggests the development of adversarial training techniques and the implementation of security measures to prevent manipulation. Techniques such as adversarial training and differential privacy can help protect AI systems against potential attacks, ensuring that fake news detection models remain reliable even in the face of sophisticated misinformation campaigns. In addition to security concerns, the real-time detection of fake news on social media platforms remains a significant challenge [14], [26], [84]. The rapid spread of misinformation requires efficient systems capable of identifying fake news as it spreads. To address this issue, our study

proposes the development of incremental learning models and optimisation algorithms for faster inference.

Online learning and streaming processing frameworks are particularly well-suited for this task, as they can handle continuous data streams and update models in real time. These approaches could significantly improve our ability to combat the spread of fake news by enabling swift identification and response to emerging misinformation. To further enhance the effectiveness of fake news detection systems, user profiling [7], [31] and credibility assessment [44], [88] represent another area where future research is needed. MODELS that analyse user behaviour, writing styles, and social network interactions could help identify potential sources of fake news. Graph Neural Networks (GNNs) are particularly promising for user profiling, as they can effectively model the complex relationships within social networks. For credibility assessment, attention-based neural networks with user embeddings can help combine user features, sentiment analysis, and semantic relationships to improve the evaluation of online conversations and news sources. Despite these advancements, the lack of diverse, multilingual, and publicly available benchmark datasets [1], [86], which hinders progress in automated media bias and fake news detection.

Creating and maintaining comprehensive datasets that cover various domains and types of fake news is crucial for better model evaluation and comparisons. Cross-lingual Transformers and data augmentation techniques can help develop these datasets and create models that perform well across different languages and contexts. Moreover, the challenge of limited labelled data [36], [103] is a persistent issue in fake news detection research. Obtaining large annotated datasets for training models is often time-consuming and expensive. To address this gap, we suggest exploring semi-supervised learning techniques and data augmentation methods. Few-shot learning and advanced data augmentation techniques can help maximise the use of available data and reduce the need for large annotated datasets, making it easier to develop effective fake news detection models with limited resources. As we work to overcome these challenges, it is important to consider the interpretability and usability of fake news detection systems [33], [91] that are crucial for their practical implementation. Machine learning solutions must be made more interpretable and usable in constrained user environments. Techniques like LIME (Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) can help make complex AI models more understandable to users, increasing the trust and adoption of these systems.

Finally, the development of cross-domain and multilingual models [94], [100] is essential for creating robust fake news detection systems that can operate effectively various contexts and languages. Approaches such as Multilingual BERT and XLM-RoBERTa show promise in this area, as they can handle multiple languages and adapt to different domains. In conclusion, addressing these research gaps requires a multifaceted approach that combines advanced machine learning

techniques, multimodal analysis, and a focus on real-world applicability. By leveraging the suggested techniques and models for each identified gap, researchers and practitioners can make significant strides in developing more effective, robust, and practical solutions to combat the spread of misinformation. As the landscape of fake news continues to evolve, it is crucial that our detection methods evolve alongside it, incorporating new technologies and approaches to stay ahead of increasingly sophisticated misinformation campaigns.

F. TRENDS AND PATTERNS IN THE LITERATURE

The literature on fake news detection research has undergone significant transformations in recent years, characterised by several interrelated trends and patterns. This review explores the evolution of this field, highlighting key developments and emerging directions. A pivotal trend is the shift from traditional ML to deep learning, with increasing reliance on transformer-based models like BERT. Early studies (e.g., [7], [69]) relied heavily on hand-crafted features and traditional ML algorithms such as Support Vector Machines (SVM) and Random Forests. In contrast, recent research (e.g., [96], [101]) demonstrates a strong preference for deep learning models, driven by their superior performance on large datasets. For instance, [101] reported an unprecedented accuracy rate of 99.96% using BERT, significantly outperforming the traditional methods.

This transition to deep learning is accompanied by an increasing sophistication in feature extraction techniques. While early studies focused on basic linguistic features and simple sentiment analysis, recent research (e.g., [15], [34]) has explored advanced methods such as Zipf's exponent and Multifractal Detrended Fluctuation Analysis (MFDFA) indices. These techniques enable a more nuanced analysis of the text structure and complexity. Furthermore, there is growing interest in multimodal approaches that integrate textual, visual, and social network information (e.g. [41], [43], [76]), which reflects the multifaceted nature of online misinformation.

The field has also witnessed an increased emphasis on transfer learning and cross-domain applicability, addressing the perennial challenge of limited labelled data in new domains and languages. References [81] and [100] demonstrate the effectiveness of transfer learning in improving fake news detection in diverse contexts. This trend is complemented by integrating the social context and user behaviour into detection models. Recent studies (e.g., [43], [44]) incorporate user behaviour patterns, social network structures, and information propagation dynamics alongside textual features, recognising that fake news detection extends beyond content analysis to understanding the social ecosystem of information spread.

As model complexity increases, researchers are prioritising XAI to ensure transparency in detection systems, which is particularly crucial for applications with significant societal impact. References [25] and [44] explored techniques to make deep learning models more interpretable, addressing

TABLE 9. Recent research gaps and proposed techniques and models.

Category	Description	Suggested Solution	Proposed Technique/Model	Suitable Dataset	Ref.
Accuracy and Performance	Existing methods struggle to achieve high accuracy in fake news detection, particularly in complex scenarios.	Develop hybrid models combining linguistic features, semantic analysis, and machine learning algorithms to improve detection accuracy.	BERT-based models with attention mechanisms	LIAR dataset, FakeNewsNet	[21], [23], [72], [80]
Multimodal Analysis	Limited exploration of text and image analysis in fake news detection.	Implement Stylometric and Semantic similarity-based multimodal detection (SSM) to capture the similarity of diverse modal data in multimodal news.	Vision-Language Pre-training (VLP) models	Fakeddit, MultiModal FakeNews Dataset	[41]
Security and Robustness	Potential security issues with explainable AI (xAI) in detection systems may enable successful adversarial attacks.	Enhance xAI systems with adversarial training and security measures to prevent manipulation.	Adversarial Training and Differential Privacy	FEVER dataset with adversarial examples	[25]
Real-time Detection	Lack of efficient real-time detection for FND on social media.	Develop incremental learning models and optimize algorithms for faster inference to prevent the rapid spread of FN.	Online Learning and Stream Processing frameworks	Twitter-based fake news datasets (e.g., PHEME dataset)	[14], [26], [84]
Feature Selection and Dimensionality	Challenges in selecting relevant features and managing high-dimensional input data for FND.	Implement TF-IDF vectors and explore dimensionality reduction techniques.	AutoML and Neural Architecture Search (NAS)	WELFake dataset	[19], [85]
Benchmark Datasets	Lack of diverse, multilingual datasets for fake news detection.	Create comprehensive, multilingual datasets for improved evaluation and comparison.	Cross-lingual Transformers and Data Augmentation	MLMA dataset, XFake dataset	[1], [86]
Content Moderation	Need for globally deployable automated content moderation systems that address bias and lack of transparency.	Develop context-driven decision-making algorithms and use explainable AI for transparency.	Reinforcement Learning for Content Moderation	Reddit moderation datasets, YouTube comments datasets	[2], [77]
Unsupervised and Transfer Learning	Limited use of unsupervised models and transfer learning techniques.	Investigate unsupervised learning for anomaly detection and domain-adaptive transfer learning methods.	Self-supervised Learning and Meta-learning	Large unlabeled news corpora, NELA-GT-2018	[28], [54]
Computational Efficiency and Privacy	High computational demands and privacy issues in detection methods.	Implement federated learning models to ensure data privacy and explore efficient model architectures.	Federated Learning and Model Compression	Distributed versions of FakeNewsNet or LIAR datasets	[5], [93]
User Profiling	Limited research on profiling approaches to identify fake news spreaders.	Develop models analyzing user behaviour, writing style, and social interactions to identify potential sources of fake news.	Graph Neural Networks (GNNs) for User Profiling	PAN-CLEF datasets for author profiling	[7], [31]
Adversarial Examples	Lack of research on generating and defending against adversarial examples.	Explore techniques for creating adversarial examples and develop robust models to withstand attacks.	Generative Adversarial Networks (GANs) for Adversarial Examples	FEVER dataset with GAN-generated adversarial examples	[18], [87]
Context-aware Detection	Insufficient consideration of contextual information, such as geographical and temporal factors, in fake news detection.	Incorporate geo-located data and temporal information into fake news detection models to improve context-aware classification.	Spatio-Temporal Neural Networks	GeoBurst+ dataset, Twitter event detection datasets	[16], [73]
Credibility Assessment	Need for effective approaches to assess the credibility of news on social networks.	Develop models that combine user features, sentiment analysis, and semantic relationships for credibility assessment.	Attention-based Neural Networks with User Embeddings	CREDBANK dataset	[77], [88]
Domain-specific Models	Lack of domain-specific models, especially for emerging topics like COVID-19.	Create specialized models and datasets for specific domains like health misinformation.	Domain-Adaptive Pre-training (DAPT)	CoAID (COVID-19 healthcare misinformation dataset), FakeCovid dataset	[8], [97]
Health Misinformation	Limited research on detecting health-related misinformation.	Develop models for evaluating the quality of medical and health-related information.	Knowledge-enhanced Neural Networks	TREC Health Misinformation Track datasets	[20], [75]
Interpretability and Usability	Lack of interpretable and user-friendly machine learning solutions.	Implement explainable AI techniques and develop user-friendly interfaces for interaction.	LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations)	FakeNewsNet with additional annotations for interpretability	[33], [91]
Cross-domain and Multilingual Models	Need for models that work across different domains and languages.	Develop robust, domain-independent models using transfer learning for multilingual detection.	Multilingual BERT and XLM-RoBERTa	MultiFC dataset, XFake dataset	[94], [100]
Multimodal Analysis	Limited research combines text, image, and video analysis for fake news detection.	Develop integrated models for analyzing multiple modalities simultaneously.	Multimodal Transformers (e.g., VisualBERT, CLIP)	Fakeddit, MM-COVID	[76], [96]cc
Annotation Schemes	Need for fine-grained annotation schemes for disinformation detection.	Develop semi-automatic annotation systems and standardized schemes for efficient labelling.	Active Learning and Human-in-the-loop Annotation	MediaEval Verifying Multimedia Use datasets	[35], [42]
Limited Labeled Data	Challenges in obtaining large, annotated samples for model training.	Explore semi-supervised learning techniques and data augmentation methods to minimize the number of labelled samples required for accurate fake news detection.	Few-shot Learning and Data Augmentation Techniques	FakeNewsNet with limited labeled samples	[36], [103]

the “black box” problem often associated with complex neural networks. Concurrently, emerging interests in few-shot and zero-shot learning approaches (e.g., [47], [100]) aim to tackle the challenge of detecting novel types of fake news with limited training data, a critical consideration given the rapidly evolving nature of misinformation tactics.

Dataset expansion and diversification represent another significant trend, with recent studies (e.g., [45], [101]) utilizing larger, more diverse datasets to test model generalization across multiple languages and domains. This trend towards comprehensive datasets is coupled with a growing focus on real-time detection capabilities, driven by the need to combat the rapid spread of misinformation on social media platforms. The increasing complexity of fake news detection has spurred interdisciplinary approaches, synthesising insights from linguistics, psychology, network science, and computer science to develop more holistic detection strategies.

Finally, there is a growing awareness of ethical considerations and potential biases in fake news detection systems. Recent literature has addressed the critical issues of fairness, accountability, and transparency in model development and deployment. Researchers are also focusing on developing models that are robust against adversarial attacks and manipulation, recognising the potential for malicious actors to exploit vulnerabilities in detection systems.

These interconnected trends underscore the dynamic and rapidly evolving nature of fake news detection research in recent years. They point towards future directions that emphasise more robust, generalisable, and ethically sound approaches to combating misinformation in the digital age, highlighting the need for continued innovation and interdisciplinary collaboration in this critical field.

G. CHALLENGES AND FUTURE DIRECTIONS IN FAKE NEWS DETECTION

The rapid proliferation of fake news on social media platforms necessitates the development of sophisticated detection systems. However, as the field of fake news detection advances, researchers and practitioners face a complex array of challenges that significantly affect the effectiveness and applicability of these systems. These challenges span multiple dimensions, including data-related issues, computational constraints, interpretability concerns, and ethical considerations. Understanding and addressing these challenges is crucial for developing robust, efficient, and trustworthy fake news detection methods. This section delves into six key categories of challenges: (1) Data-Related Challenges, highlighting issues with dataset quality and availability; (2) Computational and Architectural Challenges, focusing on the trade-offs between model complexity and efficiency; (3) Interpretability and Explainability Challenges, addressing the need for transparent AI systems; (4) Adversarial and Ethical Challenges, examining security vulnerabilities and fairness concerns; (5) Generalisability and Adaptability Challenges, exploring the difficulties in creating universally applicable detection systems; and (6) Bias in Fake News

Research. By systematically analysing these challenges and proposing potential solutions, we aim to provide a comprehensive overview of the current limitations of fake news detection and chart a course for future research directions.

1) CHALLENGES

a: DATA-RELATED CHALLENGES

The field of fake news detection continues to face significant data-related challenges that directly impact the performance, fairness, and generalisability of models. One of the most pressing issues is the scarcity of high-quality, diverse, and large-scale datasets, which are essential for training robust and adaptable models [9], [40], [90]. This scarcity is particularly evident in emerging topics and low-resource languages, where data availability is limited, thereby constraining the development of globally effective detection systems [19], [70]. Additionally, class imbalances in the available datasets pose a significant hurdle, often leading to biased learning and reduced predictive performance [85], [89]. As demonstrated by [76], the widely used Fakeddit dataset suffers from severe underrepresentation of categories such as satire, misleading content, and imposter news, which negatively affects the effectiveness of the model in fine-grained fake news classification. Moreover, the COVID-19 infodemic highlighted the difficulty of rapidly collecting and labelling data for novel misinformation topics [9], as noted in [27], who found that their dataset, although augmented via back-translation, remained limited in scope and diversity. Other reviewed studies have pointed out dataset limitations that affect generalisability. For example, [89] highlighted the linguistic and cultural limitations of the PHEM dataset, which hindered its generalisability across diverse populations and platforms. Reference [81] emphasised that many existing datasets, such as LIAR-PLUS and Covid-Stance, are either outdated or domain-specific, reducing their value for cross-domain transfer learning and real-world adaptability.

To address these challenges, future research should prioritise the development of comprehensive, multilingual, and dynamically updated datasets [19], [40], [70], [91]. In addition, advanced data augmentation techniques [46], [78]. Fostering collaborative initiatives for dataset sharing and annotation could significantly enhance the robustness and applicability of fake news detection models across different contexts and languages.

b: COMPUTATIONAL AND ARCHITECTURAL CHALLENGES

The computational complexity of advanced fake news detection models, particularly deep learning architectures, presents significant challenges for real-time detection and deployment on resource-constrained devices [8], [45]. This complexity often results in a trade-off between model sophistication and computational efficiency, impacting the ability to rapidly respond to the dynamic nature of social media content [28], [68]. For example, transformer-based models such as BERT, while highly accurate, may be too computationally intensive

for real-time applications on large-scale social media platforms [45], [101]. To overcome these challenges, researchers should explore model compression techniques, such as knowledge distillation or quantisation, to create more efficient versions of high-performance models. Additionally, developing specialised architectures optimised for fake news detection, rather than relying solely on general-purpose language models, could improve both efficiency and effectiveness of the models.

c: INTERPRETABILITY AND EXPLAINABILITY CHALLENGES

The “black box” nature of the complex neural networks used in fake news detection hinders interpretability, making it difficult to understand the reasoning behind the classifications [25], [44], [101]. This lack of transparency can erode trust in detection systems and limit their practical applicability, especially in sensitive domains such as politics or health information. To address this, researchers should prioritise developing XAI techniques specifically tailored for fake news detection. This could involve integrating attention mechanisms that highlight important features, developing modular architectures that allow for step-by-step reasoning, or creating hybrid models that combine the strengths of interpretable machine learning techniques with deep learning. Additionally, conducting user studies to assess the effectiveness of different explanation methods in enhancing human understanding and trust in model decisions is crucial [33].

d: ADVERSARIAL AND ETHICAL CHALLENGES

The vulnerability of fake news detection systems to adversarial attacks poses significant security risks [25], [46]. Malicious actors can exploit these vulnerabilities to bypass detection systems, undermining their effectiveness. Furthermore, ethical considerations surrounding the use of user data and potential biases in pre-trained models present ongoing challenges [45], [77], [100]. These issues not only affect the performance of detection systems but also raise important questions regarding fairness and privacy. To address these challenges, researchers should focus on developing adversarially robust models using techniques such as adversarial training or certified robustness. Incorporating fairness constraints and privacy-preserving techniques (e.g. federated learning) into model development processes can help address ethical concerns. Regular audits of detection systems for biases and potential misuse should be conducted.

e: GENERALIZABILITY AND ADAPTABILITY CHALLENGES

Fake news detection models often struggle to generalise across different domains, languages, and platforms [19], [80], [94], [95], [100]. This limitation is particularly problematic given the global nature of misinformation and the rapid evolution of fake news tactics. Capturing domain-specific yet language-independent features remains a significant hurdle in developing universally applicable detection systems [94], [100]. To enhance generalisability, researchers should explore advanced transfer learning techniques that can

effectively adapt pre-trained models to new domains with minimal fine-tuning. The development of multilingual and cross-domain benchmarks would also facilitate the evaluation and improvement of model generalisability. Additionally, investigating continual learning approaches could enable models to adapt to new patterns of misinformation over time, without catastrophically forgetting previously learned information.

f: IMPORTANT NOTE ON BIAS IN FAKE NEWS RESEARCH

As the field of fake news detection continues to mature, it is imperative to address a persistent and foundational issue: the presence of bias in the research itself. Despite the surge in advanced methodologies, particularly those leveraging stance detection and subjectivity analysis using transformer models [49], the literature often overlooks the subtle propagation of these biases. For example, [49] proposed a modular architecture for fake news assessment through subjectivity indicators and stance responses; however, the subjectivity ground truths are often drawn from datasets that reflect institutional biases rooted in politically affiliated fact-checking agencies. This raises concerns about a domino effect, where biased datasets underpin models which are later reused, compounding the initial distortions.

One illustrative case is the reliance on sources such as [104] in the widely referenced [49], where the foundational annotations are based on fact-checkers whose impartiality is contested. This phenomenon reflects what has been termed a “failure to rigorously and neutrally identify or even characterize bias.” Similarly, the media bias categorisation proposed [105], while innovative in classifying biases into party, region, ideology, gender, and front-runner, has embedded these implicit assumptions. Their inclusion of gender bias as a standalone category, despite statistical evidence showing only 27% female representation in national legislatures globally as of 2024 [106], underscores how uncontextualized indicators may lead to skewed interpretations of the data. As the Congressional Research Service (CRS) report highlights, such an imbalance in representation reflects societal structures rather than editorial slant, suggesting that without proper nuance, fairness assessments risk mischaracterizing demographic realities as media bias.

This layered problem has been further explored in recent reviews, such as [1], which stress that media bias is not monolithic but spans intentional and unintentional forms, including spin and ideology bias, gatekeeping, and statement bias. Most importantly, the review notes a critical research gap: fake news studies rarely self-audit for bias or discuss the limitations of their training data. Given that cognitive biases such as confirmation bias and naïve realism are prevalent among both media consumers and researchers, future work must prioritise bias awareness not only in model development but also in dataset construction and evaluation metrics. Only by critically engaging with these challenges can the field avoid undermining its credibility and instead strengthen its role in defending democratic discourse.

2) FUTURE WORKS

The promising directions for the future of fake news detection research present a multifaceted landscape of challenges and opportunities that must be addressed. Promising avenues include the exploration of continual learning methodologies to adapt to evolving misinformation patterns, the development of domain-specific pre-trained models for news articles, and the creation of multilingual and multimodal approaches to address the cross-lingual and multi-format nature of fake news propagations. Concurrently, the field would benefit from the development of larger, more diverse, and dynamically updated benchmark datasets and the expansion of research to encompass a broader range of languages and cultural contexts. Ethical considerations, particularly those surrounding the use of user data in detection systems and the delicate balance between leveraging valuable contextual information and protecting user privacy must also be addressed. The complexity of these challenges underscores the necessity of interdisciplinary collaboration, integrating insights from computer science, linguistics, and social sciences to formulate comprehensive strategies for the detection and mitigation of fake news. By pursuing these diverse research directions and fostering collaborative efforts, the field can work towards developing more robust, efficient, and ethical sound systems capable of effectively combating the ever-evolving landscape of online misinformation. To guide future research efforts, we propose the following prioritized roadmap:

- 1) **Short-term priorities** should focus on addressing data-related challenges, particularly the development of multilingual large-scale datasets and methods for handling class imbalances. These form the foundation for all the subsequent improvements.
- 2) **Mid-term goals** include addressing interpretability and ethical challenges by developing XAI models and integrating privacy-preserving mechanisms.
- 3) **Long-term objectives** should target generalisability and adaptability, aiming to create cross-lingual and cross-domain models capable of continual learning and resistant to adversarial manipulation.

This roadmap emphasises an incremental yet integrated approach, where foundational improvements in data and transparency pave the way for robust, adaptive, and ethically aligned fake news detection systems.

As part of the long-term objectives focused on adaptability and generalisability, few-shot and zero-shot learning approaches represent particularly promising directions. To support future research in this area, we propose the following research questions:

- 1) How can few-shot learning frameworks be optimised to detect misinformation related to emerging topics using minimal supervision?
- 2) To what extent can zero-shot learning generalise across languages, platforms, and news domains without requiring fine-tuning?

- 3) What prompt engineering strategies or embedding techniques can improve the interpretability and performance of zero-shot models in fake news detection tasks?

Addressing these questions could accelerate the development of lightweight, generalisable systems capable of responding to the rapidly evolving nature of misinformation.

These challenges underscore the importance of acknowledging the limitations of this review, as outlined in the following section.

V. STUDY LIMITATION

While this systematic literature review offers valuable insights into recent advancements in fake news detection using machine and deep learning approaches, several limitations should be acknowledged to contextualise its findings.

First, the exclusive reliance on the Web of Science (WoS) database, although selected for its indexing rigor and peer-reviewed coverage, may have limited the review's comprehensiveness. Broader academic databases, such as Scopus or Google Scholar, which index a wider array of journals, grey literature, and preprints, were not included. Future reviews may benefit from incorporating multiple databases to enhance the retrieval scope and minimise potential omissions.

Second, the search strategy employed a broad string to capture varied terminology used in the literature, including terms like "misinformation" and "disinformation." While this approach ensured inclusivity, it may have introduced irrelevant studies. To maintain thematic precision, strict inclusion criteria were applied based on a clearly defined concept of fake news. Nonetheless, narrower search strings could have improved the initial result relevance, albeit at the risk of excluding valid contributions owing to inconsistent terminology usage across the field.

Third, the review was restricted to studies that used English-language datasets. Although this decision was made to ensure analytical consistency and comparability, it may limit the generalisability of our findings to different linguistic and cultural contexts. Given the global nature of fake news, future research should explore multilingual corpora and cross-lingual model evaluations to enhance inclusivity and broaden their practical applicability. Additionally, a formal statistical meta-analysis was not conducted due to the high heterogeneity among the reviewed studies. The wide variation in model types, evaluation protocols, and datasets, ranging from domain-specific corpora to multimodal and multilingual resources, renders statistical aggregation methodologically inappropriate. Instead, performance comparisons were synthesised qualitatively and presented in a structured comparative table to ensure meaningful interpretation within the context of each study. Although this review is centred on model-centric research questions, broader ethical concerns, such as false positives, algorithmic bias, censorship risks, and implications for free speech, are beyond its defined scope but remain critical areas for future interdisciplinary investigation. Additionally,

the use of user metadata and social network information introduces important privacy concerns. Models that leverage behavioural signals, profile attributes, or interaction data may inadvertently compromise user privacy if they are not carefully governed. These issues underscore the need for transparent data governance, informed consent protocols, and compliance with data protection regulations, which should be further addressed in future studies combining technical development with ethical oversight.

VI. CONCLUSION

This systematic literature review examined recent advancements in machine- and deep-learning approaches for detecting fake news on social media platforms. Adhering to the PRISMA guidelines, we analysed 90 peer-reviewed papers published between 2020 and 2024 to identify emerging trends, technical contributions, and prevailing challenges in the field. Our findings confirm that advanced deep learning architectures, particularly Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models such as BERT, consistently outperform traditional machine learning methods across various fake news detection benchmarks. Notably, a BERT-based model achieved nearly 99% accuracy on the ISOT, Fake-or-Real, and D3 datasets, and up to 99.9% accuracy on the Kaggle dataset. The GANM model, which integrates textual and social contextual features, also demonstrated strong performance on the FakeNewsNet dataset. Despite their superior accuracy and generalisability, these models often involve trade-offs in terms of computational efficiency and scalability, which can impact real-time deployment on large-scale social media platforms. These results are further detailed in the “Evaluation Metrics” column of Table 6, which includes accuracy (Acc), precision (P), recall (R), and F1-score (F1) for each reviewed study, offering a comprehensive basis for comparing effectiveness across different approaches. Transfer learning has emerged as a key enabler for adapting models to different domains and low-resource languages (LRLs). Moreover, few-shot and zero-shot learning approaches show promise for detecting emerging misinformation with minimal supervision, addressing the “cold start” problem in dynamic information environments. However, issues such as scalability, real-time performance, and generalisability across domains remain challenges. Additionally, the adoption of explainable artificial intelligence (xAI) methods, such as SHAP, co-attention networks, and prompt-based reasoning, is vital for improving interpretability and fostering user trust. This review also recognises several methodological and scope-related limitations discussed in Section V, which should be considered when interpreting the results.

Despite the breadth of our analysis, this review has several limitations. First, it exclusively considered English-language studies indexed in the Web of Science database, which may limit the cultural and linguistic diversity of the findings. Second, few-shot and zero-shot learning approaches, while highlighted as promising, are still in the early stages

of development, and empirical research remains limited. Third, our review identified a gap in user-centred evaluation of interpretability; few studies assessed how explanations affect user trust or decision-making in real-world settings.

To address these gaps, future work should explore multilingual and cross-cultural datasets, evaluate fake news detection methods on diverse platforms, and incorporate interdisciplinary perspectives, especially from the social sciences and human-computer interaction. Advancing explainability through techniques such as co-attention mechanisms, SHAP, and prompt engineering will be vital, as will adversarial robustness for explanation generation. Furthermore, building dynamically adaptive systems that consider temporal patterns and evolving misinformation tactics will be key to ensuring their real-world relevance.

In conclusion, the future of fake news detection lies in the development of robust, adaptive, and ethically sound systems that balance accuracy, transparency, and inclusivity. By leveraging the trends identified in this review and addressing its limitations, researchers and practitioners can contribute to the creation of a more resilient and trustworthy digital information ecosystem.

DECLARATIONS

The authors declare no known financial or personal relationships that could have influenced this work and confirm that no conflicts of interest or competing interests are associated with this manuscript.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this manuscript, the authors used an AI tool to edit the English language. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

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